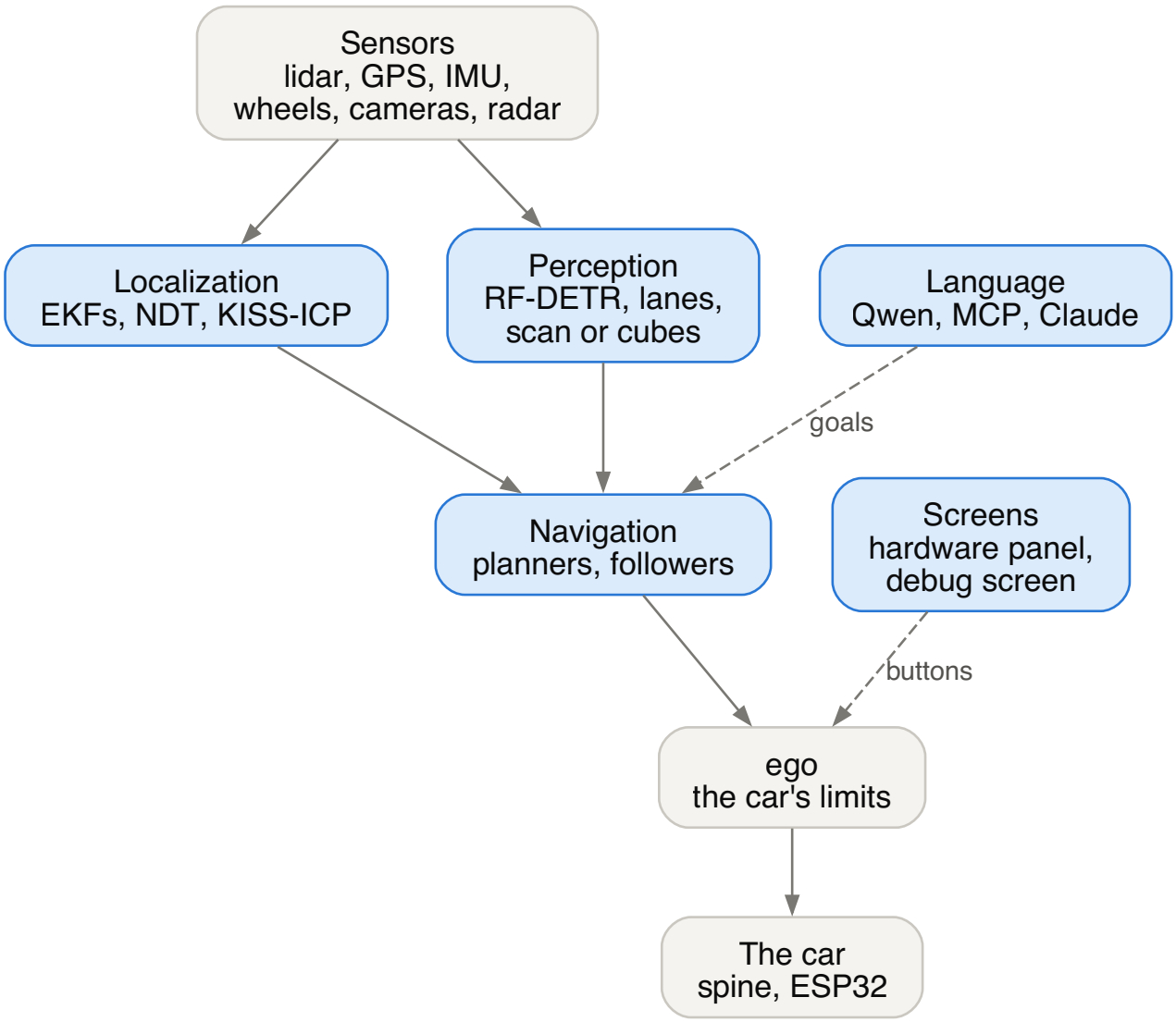
CARLETON UNIVERSITY · AUTONOMOUS VEHICLE CAPSTONE

AAV 0.2: the guide

What each part of release 0.2 does, how to run it, and how far to trust it. It carries on from What's new (22 October to 1 November), which stays as history.

The short version

- **0.2 = sim-complete, car-ready.** Everything runs in Gazebo and every test passes. **Nothing has moved the real car yet**: that's 0.3.
- **Screens:** the HUD, the map, the debug screen, the hardware panel (with the steering calibration) and the cockpit; Initial D red and black on all of them.
- **Safety:** the joystick-alive check, and in the Rust ESP32 firmware the heartbeat brake and the state machine.
- **Car day:** Day A destroys nothing; Day B upgrades, once a restore is proven.
- **Commands:** 45, in groups. New here? Start at Your first hour.
- **Next:** the 4 Nov meeting, then car day A. The plan.



The whole stack on one page. Blue: built from 22 October on. Grey: there before.

Inside

How to read this page	4	The state machine	22
Contents	4	Localization: where am I?	24
Your first hour	5	Why don't the two EKFs talk?	27
0.2 in one page	6	Perception: what's around?	29
The plan	7	Planning: drawing the path	31
The defaults	8	Following: steering along the path	34
The screens	11	Safety: the layers that stop the car	36
The joystick and the joystick-alive check	20	At the car	38
The Rust ESP32 firmware	20	Tools	40

Which parts are Python, and is that OK? 44
Kept for 0.3 and later 46
How solid is 0.2? 47
The checks run for this page 47

How to read this page

Each part has one small table:

Row	Says
What it does	in plain words
Run it	the exact command, from the repo folder
Tips	what trips people up
How solid	one of the words below

Word	Means
sim	works in Gazebo (or the bench), tested
built	built and tested on its own, not yet in a full run
written	a runbook or a sheet, no code to run
trial	runs beside the default, off unless you turn it on
not on the car yet	nothing here has run on the real car

Contents

Part	Sections
Start	Your first hour · 0.2 in one page · The plan · The defaults
Screens	HUD · Map · Debug screen · Hardware panel · Steering calibration · Cockpit · Colours
Driving by hand	The joystick
Firmware	The Rust ESP32 · The state machine
How the car thinks	Localization · Why the EKFs don't talk · Perception · Planning · Following
Safety	The safety layers
At the car	car_setup.sh · Car day · Measurements · Recording

Part	Sections
Tools	Foxglove and Lichtblick · MCP server · The commands
Honest	Which parts are Python · Kept for 0.3 · How solid is 0.2?

Your first hour

New to the team? Run these in order, on a Mac or on Linux. The setup before step 1 (Pixi, your fork): README's "Your first day".

#	Run	You should see	Time
1	`pixi install`	ROS 2 Jazzy, Gazebo, Rust: about 7 GB	the first time, minutes
2	`pixi run build`	every package built	a few minutes
3	`pixi run test`	every test green	a few minutes
4	`pixi run turtle`, then in a second terminal `pixi run keyboard --local`	a turtle that steers like the car: hold `i` to drive, `u` or `o` to turn	2 min
5	`pixi run hud --demo`, then `pixi run hw-panel --demo`	the in-car screen and the hardware panel, with pretend data (no ROS)	1 min
6	`pixi run pick`	a menu: Gazebo, the planner, the path follower, where-am-I. It prints the command and runs it	5 min
7	`pixi run sim-check`	Gazebo's road world, headless, and the whole graph checked	2 min
8	`pixi run docs`, then open `docs/html/index.html`	these pages in a browser	1 min

Tips

- Something fails? Run it lists the usual causes.

- Gazebo shows goals accepted but the car never moves: stop everything, then `pixi run fastdds shm clean`, then start again.
- "This computer only" commands (`turtle`, `sim`, `bench`...) never reach a real car on the same network. `keyboard` with no flag does: use `--local` or `--sim`.

0.2 in one page

Area	What 0.2 has	How solid
Screens	HUD, map, debug screen, hardware panel (2 pages), cockpit; Initial D on every screen	sim
Steering calibration	find the steering actuator's ends and centre: limp or jog, mark, equal limits, a lock-to-lock check	built
Joystick	the joystick-alive check: a dead pad stops the car	sim (bench 11 of 11)
Rust ESP32	the heartbeat brake (100 ms), link versions 1-3, the state machine	built (QEMU)
Localization	two EKFs (default); AMCL, NDT, NDT-EKF, KISS-ICP as trials	sim
Perception	RF-DETR Nano on the camera stream; the lidar as a flat scan (default) or cubes	sim
Planning, following	the lattice or Hybrid-A*, MPPI; trials: diffusion, RPP, our trackers	sim
Safety	e-stops, watchdogs, the output mux, the geofence	sim
Car day	Day A and Day B, the people, sign-offs, `car_setup.sh`	written; dry run only
Commands	45, grouped	done

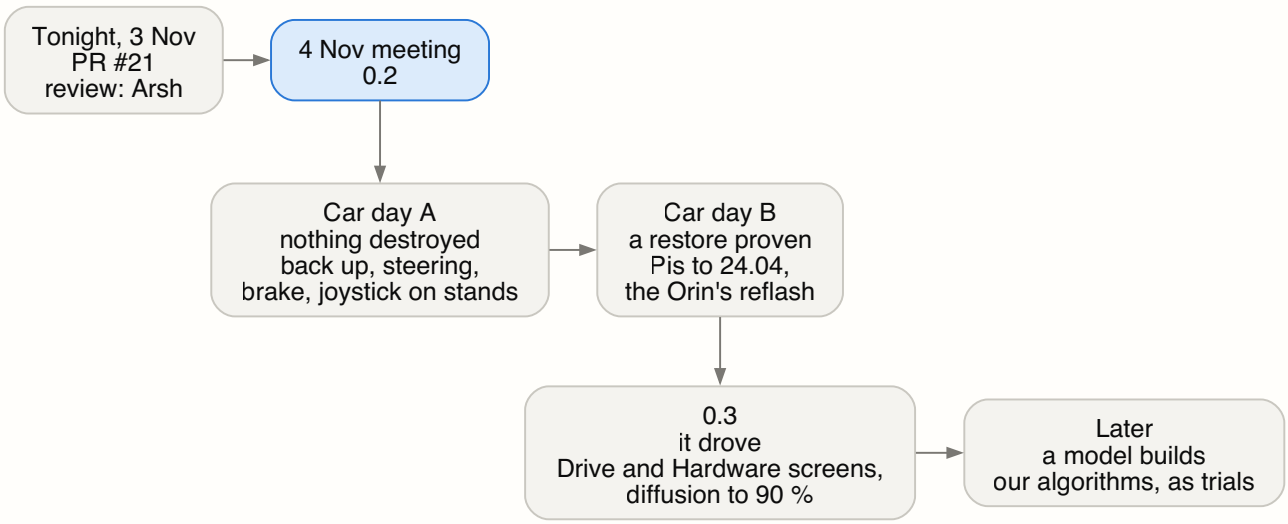
When is 0.2 done?

When every box on the 0.2 checklist is ticked. Where it stood on 2 November:

Check	State
Every test green (`pixi run test`)	✅ 1516 tests, 0 failures (2 Nov)

Check	State
The kit ready (`car-day-check`)	⬜ 11 missing: downloads and the ESP32 tools (below)
The ESP32's emulator check (`esp qemu-check`)	⬜ the ESP32 tools not installed on this Mac yet
`car_setup.sh`'s dry run, every role	✅ only the two planned refusals
The first-move gaps (TODO 74)	🔧 the joystick-alive check built, passed on the bench (2 Nov), its car test in A9b; ⬜ the throttle cap
The people named	⬜ Ben and the team
The hardware lead's answers (H1, C1, `.211`, the router)	⬜ the hardware lead

The plan



From tonight to 0.3 and after. Blue: where we are.

When	What	Who decides
Tonight, 3 Nov	PR #21 from the fork's `main`, after the full tests (TODO 91); review by Arsh. The people named, the hardware lead's OK for A8 and H1, the restore test's spare SD card (TODO 92)	Ben; Arsh reviews

When	What	Who decides
4 Nov meeting	0.2: this guide, on screen	the team
Car day A (nothing destroyed)	back up every machine; our stack beside theirs; the steering's ends; the brake on stands; the Rust ESP32, if the hardware lead agrees; the joystick on stands; walking pace; measure; record all day	the hardware lead for the hardware; an approver signs each step you can't undo
Car day B (later)	a restore proven first; the Pis to Ubuntu 24.04; the Orin's reflash, once JetPack 7.2 is confirmed	the hardware lead; the approver
0.3 = it drove	one screen per concept: **Drive** and **Hardware** (TODO 90); the hardware panel's pages; brake and throttle setting; diffusion to 90 % and on the Orin; Rust where the Orin's timing says so; the cubes and the post (TODO 93)	the team; the hardware lead for the hardware
Later	a model builds our algorithms from the recordings, each one a trial beside the default (TODO 96)	the team, after car day's data

The car day steps, one by one: Car day, the order.

The defaults

What runs when you give no switch. Every switch: Switches.

System	In Gazebo (`sim`, `demo`)	On the car (`car`)	Set in	Trials beside it
Planner	lattice; Hybrid-A* when forward only	the same; the car refuses every other planner	`vehicle.py`: `DEFAULT_PLANNER`, `default_planner()`	diffusion, smac2d, navfn, theta_star
Path follower	MPPI	MPPI	`vehicle.py`: `DEFAULT_CONTROLLER`	RPP; our trackers (`tracker:=`)

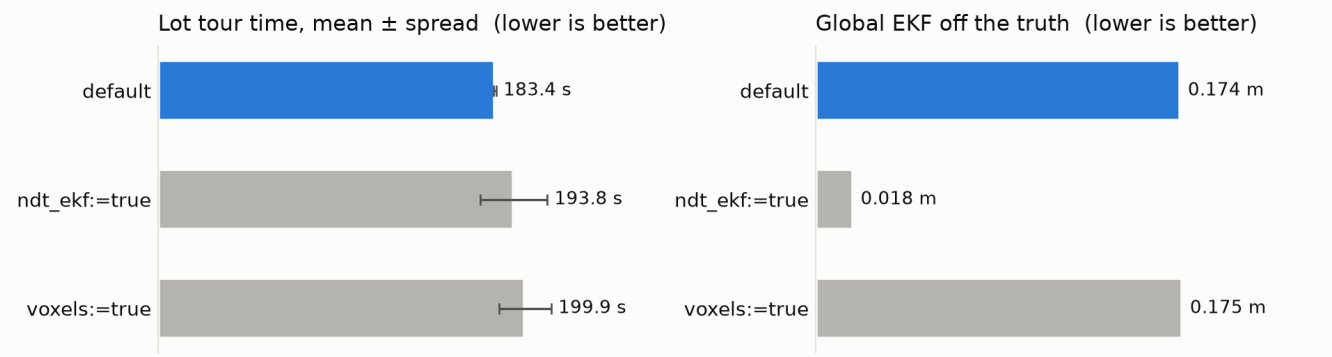
System	In Gazebo (`sim`, `demo`)	On the car (`car`)	Set in	Trials beside it
Where am I	the two EKFs: wheels, IMU, GPS	the same	`aav_localization/config/ekf_*.yaml`	AMCL, NDT, NDT-EKF, KISS-ICP
Obstacles for Nav2	the lidar as a flat scan	the same	`voxels:=false` in the launch files	the cubes (`voxels:=true`)
Speed loop	`default` (PI)	the Motor Pi bridge's, open loop (no wheel speed on the ESP32)	`speed_loop:=`; `aav_legacy/config/legacy.yaml`	`lqi`, `smooth`
Geofence	on, 30 m round the start	on	`geofence:=`, `geofence_radius:=`	
Serial link to the ESP32		**version 1** (the C++ firmware): programs locked out, only people and the actuator tests drive	`legacy.yaml`: `frame_version`	version 2 (heartbeat), 3 (calibration)
CAN bridge		Rust (`aav_spine_bridge`)	`bridges.py`: `DEFAULT`	`bridge:=python`
Screens' colours	Initial D	Initial D	each screen's yaml: `theme:`	`mocha`, `latte`

Why these (Gazebo's lot tour, reversing on, 3 runs each, 3 Nov; trials):

Candidate	Legs	Tour time	Global EKF off	Default?
today's defaults	12 of 12	**183.4 ± 0.7 s**	0.174 m	yes

Candidate	Legs	Tour time	Global EKF off	Default?
NDT into the EKF (`ndt_ekf:=true`)	12 of 12	193.8 ± 18.1 s	**0.018 m**	not yet: it needs the lot's NDT map (`maps/ndt/test_lot.pcd`), which isn't in git, so a new laptop or the road world can't start it. Next: the default of the test-lot demos, which already need that map; on the car after the lot is recorded
the cubes (`voxels:=true`)	12 of 12	199.9 ± 14.1 s	0.175 m	**no**: slower, and once in 3 runs the car drove 1.30 m into a post (`sim-check --lidar-obstacle --voxels`); to find out why
diffusion (`planner:=diffusion`)	12 of 12	208.5 ± 53.4 s (177, 179 and one 270)	-	no: 73 % of plans from the model (the goal 90 %)

The lattice is the quickest round the lot with reversing; Hybrid-A* keeps the U-turn when the car may only go forward. MPPI finished the most legs of Nav2's two. Every one of the 23 switches started and finished a leg (3 Nov).



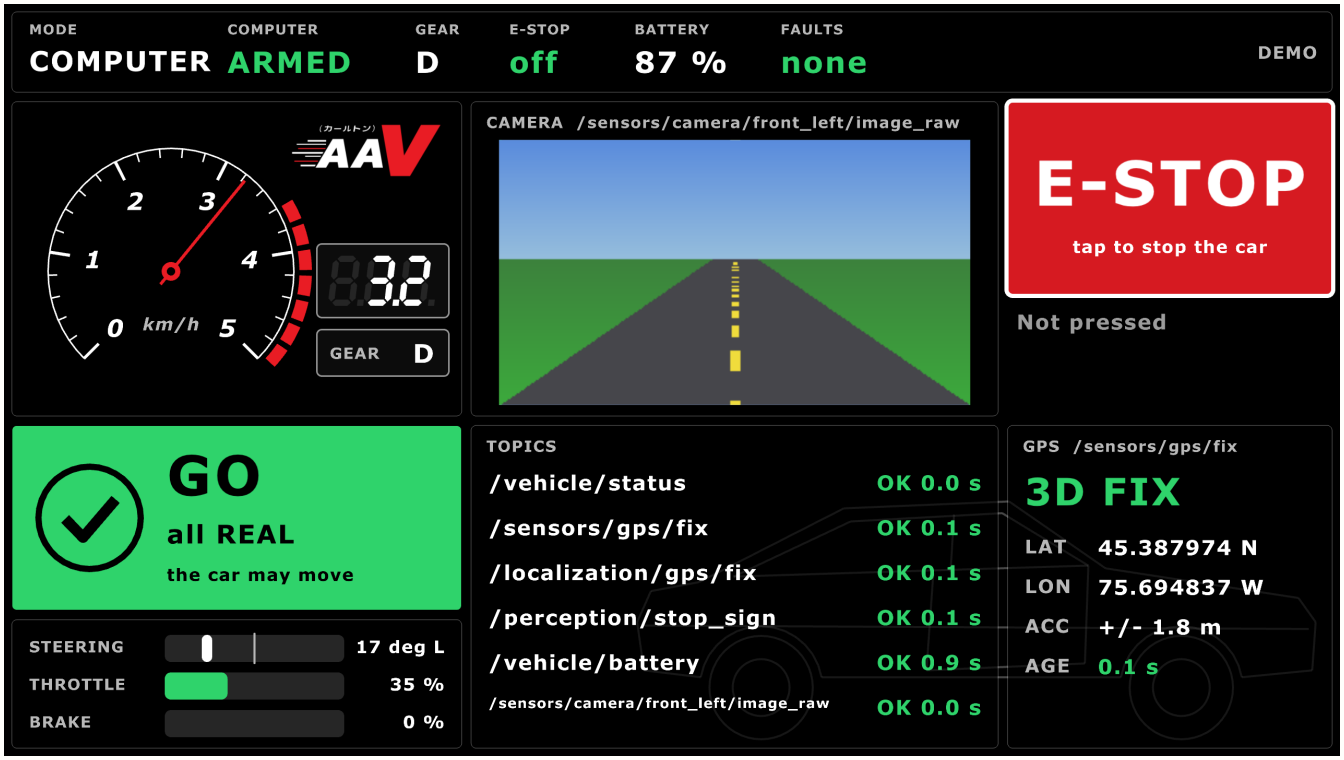
Gazebo's lot tour, 3 Nov, 3 runs each, 12 of 12 legs every run. Blue: what runs today. NDT-EKF waits for a map in git.

Today's defaults are the quickest; NDT-EKF is 10 times closer to the truth, but needs a map that isn't in git.

The screens

Screen	Where	For
HUD	the car's small screen	speed, may the car move, the camera
Map	the car's big screen	the route on Ottawa's streets
Debug screen	a laptop or the lab	what the car sees, in 3D
Hardware panel	the Orin's screen or a wired laptop	every sensor's number, simple buttons, the steering's ends
Cockpit (Lichtblick): the Drive screen	the lab or a laptop	every driving detail on one screen, and which algorithms run

The HUD

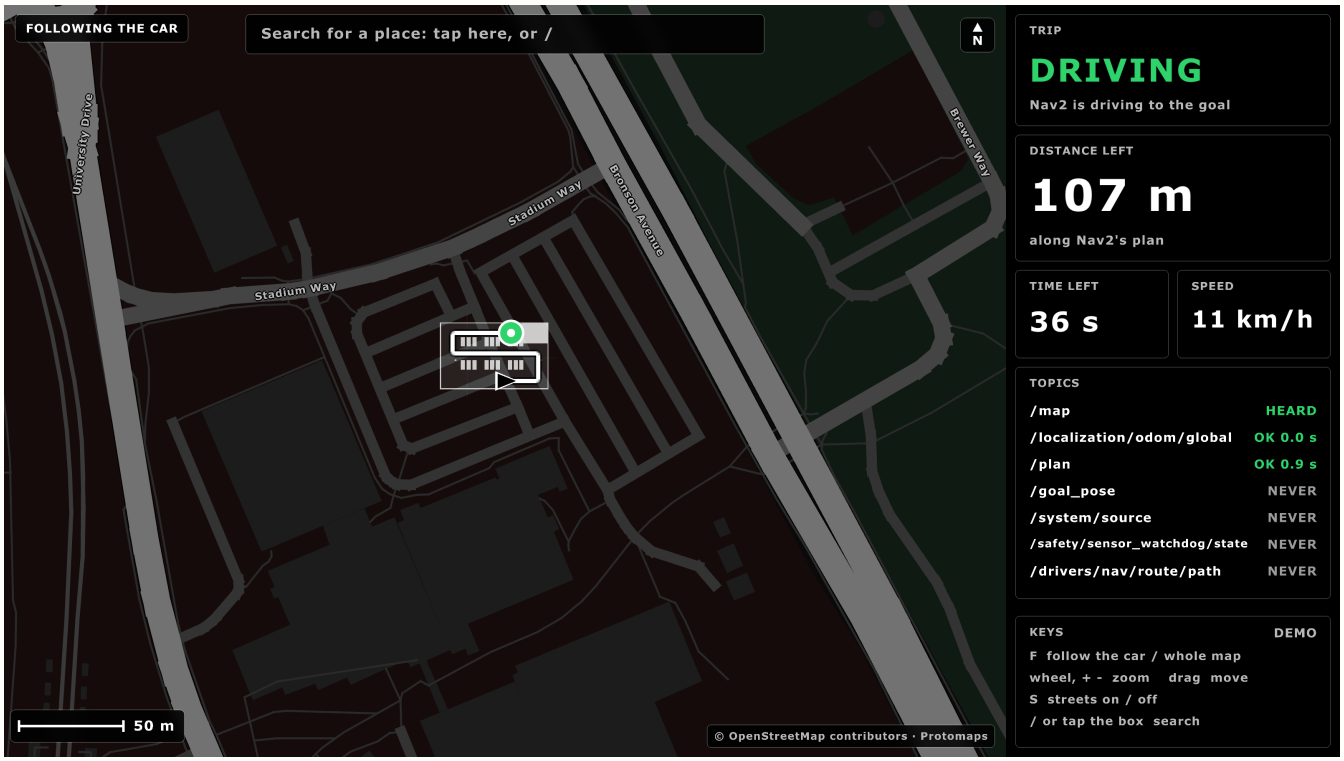


What it does	the in-car screen: speed, mode, gear, the GO / STOP panel (may the car move, and the one main reason), the camera with stop-sign and SIMULATED banners, the GPS, every topic's age, the E-STOP button
Run it	`pixi run hud --demo` (pretend data, no ROS); `pixi run hud` beside `sim`; on the car it starts at login (The two screens)

Tips	grey speed with "OLD" means `/vehicle/status` stopped. The E-STOP button only ever sends true: clear it with `pixi run clear estop`. Full screen: `fullscreen:=true monitor:=HDMI-2`
How solid	sim; Rust and raylib. Not on the car yet (its Pi still runs Humble)

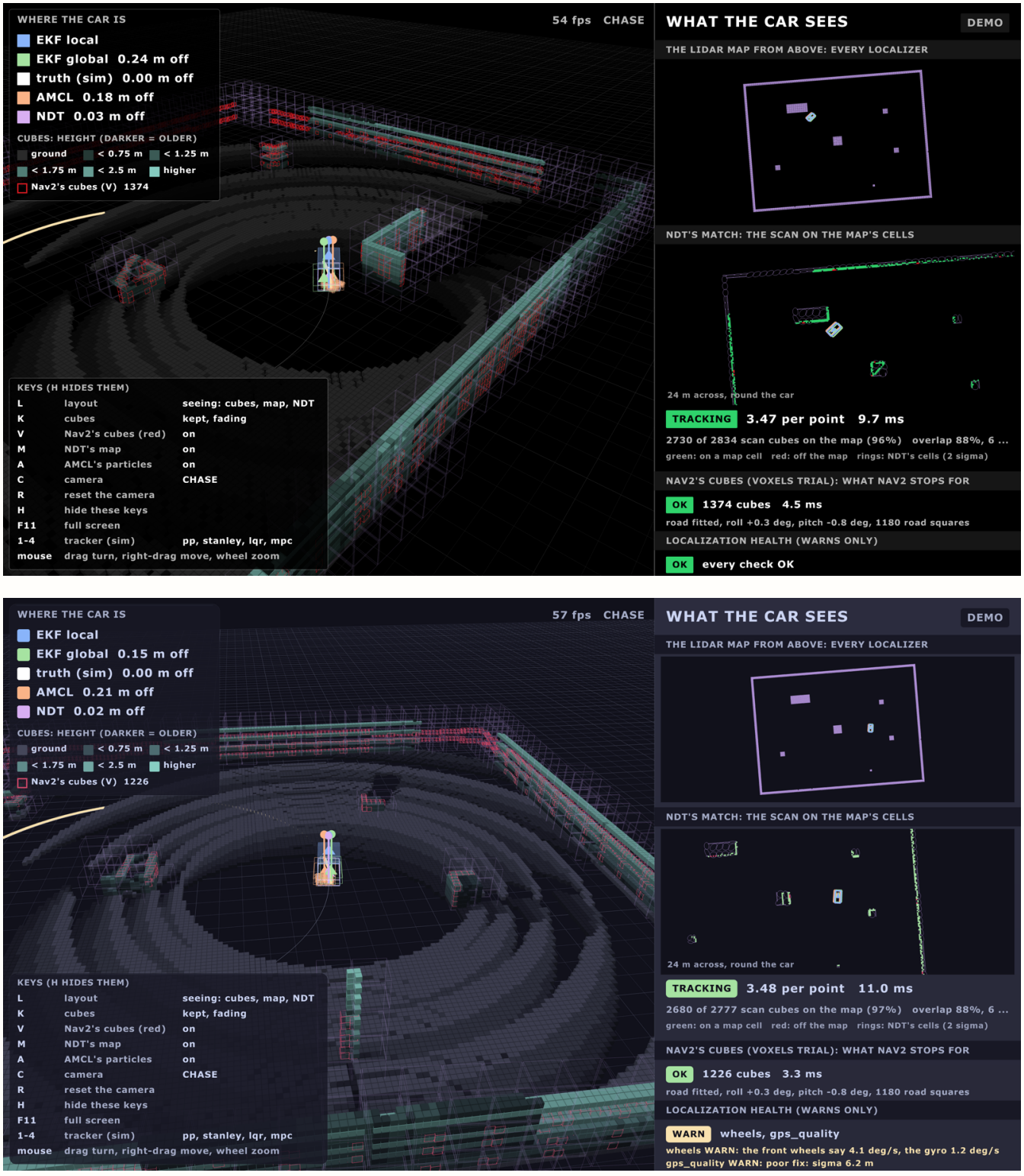
More: HUD.

The map window



What it does	the car on Nav2's map over Ottawa's streets; search for a place and drive there
Run it	`pixi run map --demo` ; with Gazebo: `pixi run sim world:=test_lot map:=true`
Tips	the streets need `pixi run get street-map` once, the search `pixi run get places`
How solid	sim. Not on the car yet

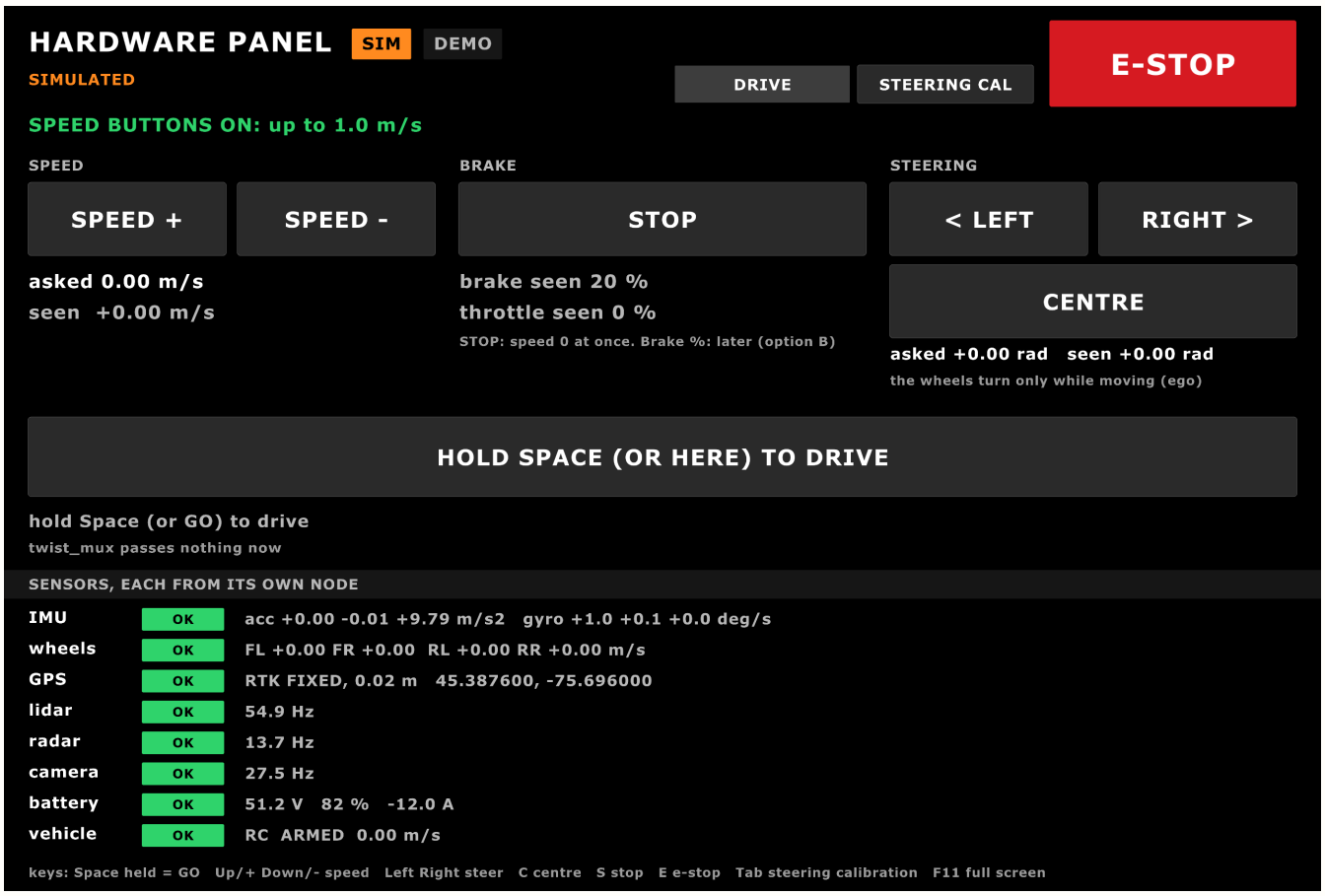
The debug screen



The debug screen, 1 Nov: Nav2's cubes in red (key V), where each localizer puts the car, and the localization health checks.

What it does	what the car sees, in 3D: the lidar's cubes, where each localizer puts the car (EKF, AMCL, NDT), Nav2's own cubes, the localization health checks
Run it	`pixi run demo --debug` (Gazebo, RViz, the HUD, the map and this); `pixi run debug-hud --demo` (no ROS)
Tips	keys 1-4 swap the path follower live (sim only). NDT shows only once its map exists: `pixi run ndt-map drive` (about 6 min)
How solid	sim

The hardware panel





The hardware panel driving Gazebo's car: every sensor's number, and the simple buttons.

What it does	the hardware at the lowest level: every sensor from its own node, and very simple buttons. The car moves **only while you hold Space**
Run it	`pixi run hw-panel --demo` ; with Gazebo: `pixi run sim world:=test_lot` , then `pixi run hw-panel` ; on stands: `pixi run hw-panel stands:=true`
Tips	run it from the Orin's screen or a wired laptop, not over WiFi. On the real car the speed buttons stay **off**, and the screen says why: no driven-wheel speed yet (TODO 49b)
How solid	sim. Not on the car yet

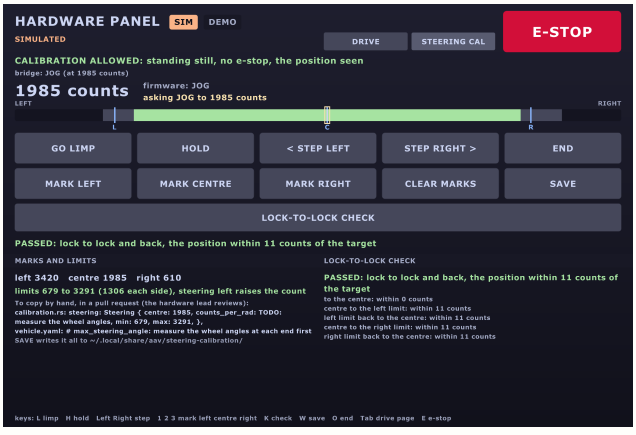
Steering: find the ends and the centre

The panel's second page, **STEERING CAL** (Tab). The steering pulls right: its centre is probably off. This page finds the steering actuator's two ends and its centre, then sets **equal limits** on both sides.

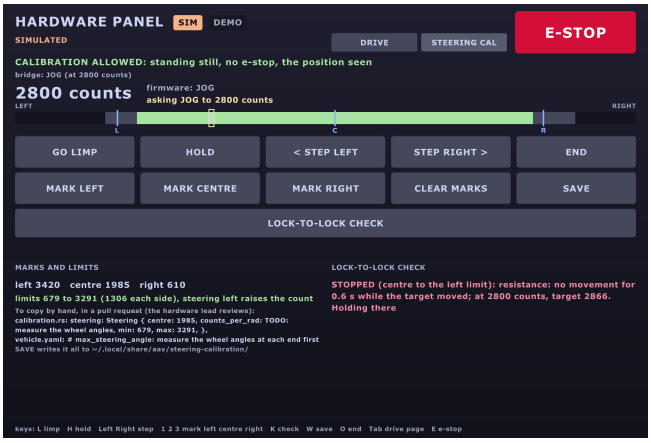
```
laptop (hardware panel)  ──►  Orin bridge  ──►  ESP32  ──►  steering actuator
     GO LIMP / STEP           link version 3     limp, or       (Jrk G2 board)
     MARK left/centre/right                      small steps
     ◄── the raw position (counts), 10 times a second ──┘
```

Step	What you do	What it does
1	**GO LIMP**	the motor lets go; you turn the wheels by hand
2	or **STEP LEFT / RIGHT**	small steps, inside today's limits
3	**MARK** left, centre, right	it works out equal limits: the smaller side less 5 %
4	**LOCK-TO-LOCK CHECK**	slowly centre → left → centre → right → centre; **stops and holds** at any resistance
5	**SAVE**	writes the numbers to a file; a person copies them in a pull request

Passed



Stopped on a stiff spot

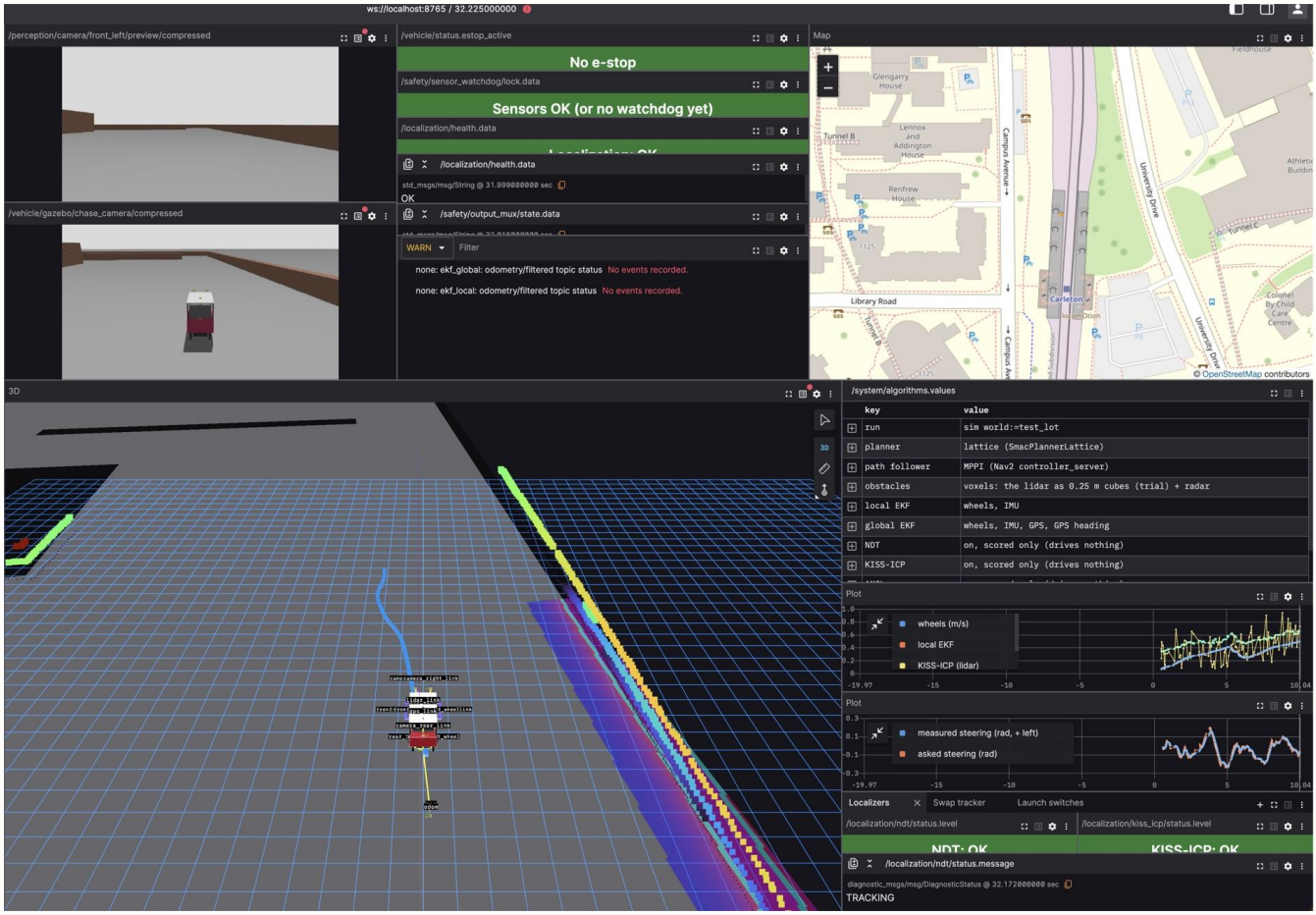


Run it	`pixi run hw-panel --demo`, then Tab. On the car: the Rust firmware, `frame_version: 3`, `uplink: true` and `steering_calibration: stands` in `legacy.yaml`
Tips	only while stopped, with the brake on; throttle held at 0. Losing the Orin's heartbeat for 100 ms ends it. It never writes the firmware or `vehicle.yaml`
How solid	built. **On car day the first way is Pololu's own program over USB** (Way A): today's C++ firmware can't do this page

More: Actuator tests.

The cockpit

The **Drive** screen: every driving detail on one screen (Ben, 3 Nov: "the gazeboo with nav2 selection with cubes and stuff all as a single screen"): both cameras, the status lights, the street map, the 3D view with the cubes, the plan and **each localizer**, the speed and steering plots, and a new **Algorithms** table: which planner, path follower and localizers this run uses. Live in Gazebo's test lot, 3 Nov:



Panel	New on 3 Nov
Algorithms	planner, path follower, obstacles, the EKFs' inputs, NDT, KISS-ICP, AMCL (`/system/algorithms`)
3D	`/map` ; NDT's and AMCL's poses; the EKFs' arrows and NDT's scan (scripts: not showing yet)
Speed plot	KISS-ICP's speed and the local EKF's beside the wheels'
Tabs	Localizers (NDT, KISS-ICP lights) · Swap tracker · Launch switches

What it does	everything for driving on one screen: Gazebo's view or the car's cameras, Nav2's goal, the cubes, the localizers, which algorithms run
Run it	`pixi run sim world:=test_lot foxglove:=true` (add `voxels:=true ndt:=true kiss_icp:=true` to see them), then in Lichtblick: Foxglove WebSocket `ws://localhost:8765` , and import `tools/foxglove/cockpit.json`
Tips	send a goal with the 3D panel's **Publish** tool, set to **Pose**. Lichtblick keeps its old copy of a layout it already has: import under a new name. On the car it is watch only. The street map needs the internet
How solid	sim, seen live 3 Nov. Not showing yet: the EKFs' arrows and NDT's scan (two layout scripts); NDT's map and KISS-ICP's cloud have no topic yet

More: Watch from a laptop. Where it's going (0.3): one **Drive** screen that takes in the debug screen's views, and the **Hardware** panel apart (TODO 90):

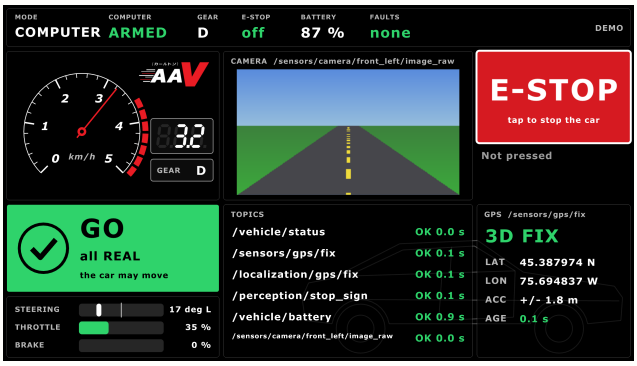
Screen	0.2	0.3
Drive (the cockpit)	driving, Nav2, cubes, localizers, algorithms	absorbs the debug screen
Hardware (the panel)	sensors, buttons, steering's ends	stays separate
HUD and **map**	in the car	stay in the car

The colours and the little robot

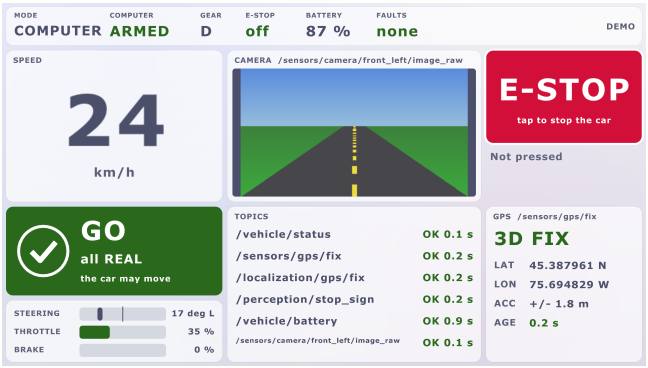
Ben, 2 Nov: "something nicer than Catppuccin". **Initial D red and black** is the default on every screen. Change it with `theme:` in the screen's yaml.

Theme	For
`initial_d` (the default)	the car, day or night: black, white, Carleton's red
`mocha`	dark and soft: night
`latte`	light: daytime
`calm`	Ben's colours from his Carleton work: **waits on Ben**

Initial D

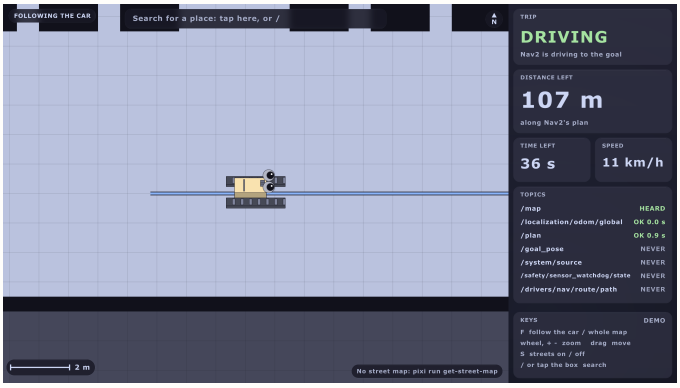


Light (Latte)

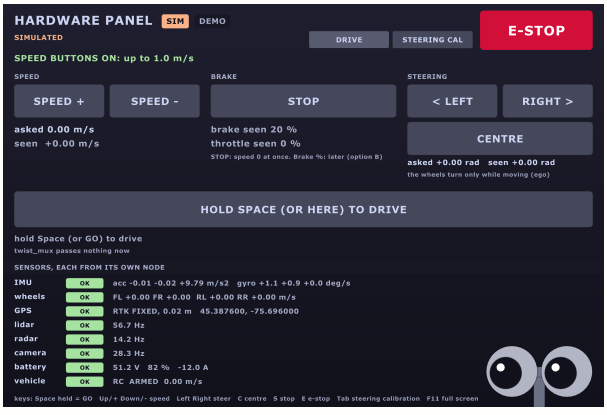


The little robot (a tribute to WALL-E): the car on the map (`car_marker: robot`), and eyes that peek after 60 s of quiet on the panel and the debug screen. Never while the car moves or anything warns. Off: `easter_egg: false`.

On the map



Peeking



The joystick and the joystick-alive check

```
joystick (USB, on the Pi) ─► joy_alive ─► teleop ─► twist_mux ─► ego ─► output mux ─►
legacy bridge ─► ESP32
                                silent 0.5 s  hold-to-   0.5 s       0.3 s   the
heartbeat   0.3 s           brakes 100 ms
                                frozen 2 s    drive      timeout     timeout
counter         stale→brake     after it stops
```

What it does	drive with the joystick, all on the Motor Pi (`machine:=motor_pi_alone`). `joy_alive` stops it when the pad goes **SILENT** (nothing for 0.5 s) or **FROZEN** (every value the same for 2 s while driving: a dead pad's receiver repeats them). It latches until you release the trigger
Run it	sim: `pixi run sim teleop:=true` with the pad plugged in. On stands: `~/aav-jazzy/aav.sh ros2 launch aav_bringup car.launch.py machine:=motor_pi_alone max_speed:=0.3`
Tips	watch `ros2 topic echo /drivers/joy/alive` : `WAITING` until the trigger is released once. A stick pinned at its edge trips FROZEN too: release, press again. Button and axis numbers: change `teleop_twist_joy.yaml` and `joy_alive.yaml` together
How solid	sim, bench-check 11 of 11. The real pad: car day A9b. **The Rust firmware only**: the C++ has no timeout

More: Car day, A9b.

The Rust ESP32 firmware

Today's car runs the 2025-26 C++ on its ESP32. **It has no command timeout**: if the Pi dies mid-command, the last throttle stays. Our Rust firmware (`boards/esp32-umod` , on `spine-core`) fixes that. It is not flashed yet: that's car day A8, with the hardware lead's OK.

The heartbeat brake

```
 Orin output mux ──counter +1 per command──► Motor Pi bridge ──copies it, 50 Hz──► ESP32
                                                          (never makes one)
counter unchanged

100 ms → brake full,

throttle 0, latched
```

What it does	the ESP32 counts a frame as a command only if the Orin's counter changed. 100 ms without a new one: brake full, throttle 0, latched until a stop with a **new** counter
Run it	`pixi run esp tools` and `pixi run esp qemu-tools` (once), then `pixi run esp qemu-check`
Tips	the Pi relays, it never makes a counter: a stuck Pi resending frames still brakes the car
How solid	built: QEMU braked 111-120 ms after the counter froze. Never on a real ESP32

The throttle cap

What it does	the throttle's top is DAC 700 of 1023, rising at most 600 a second (`spine-board calibration.rs`). The test mode caps it at 0.15
Tips	the ESP32 can't see the speed, so it can cap only the throttle, not the speed. ego's `max_speed:=0.3` caps the speed
How solid	built. **The first moves' number is open** (TODO 74b): Ben and the hardware lead decide it

Link versions

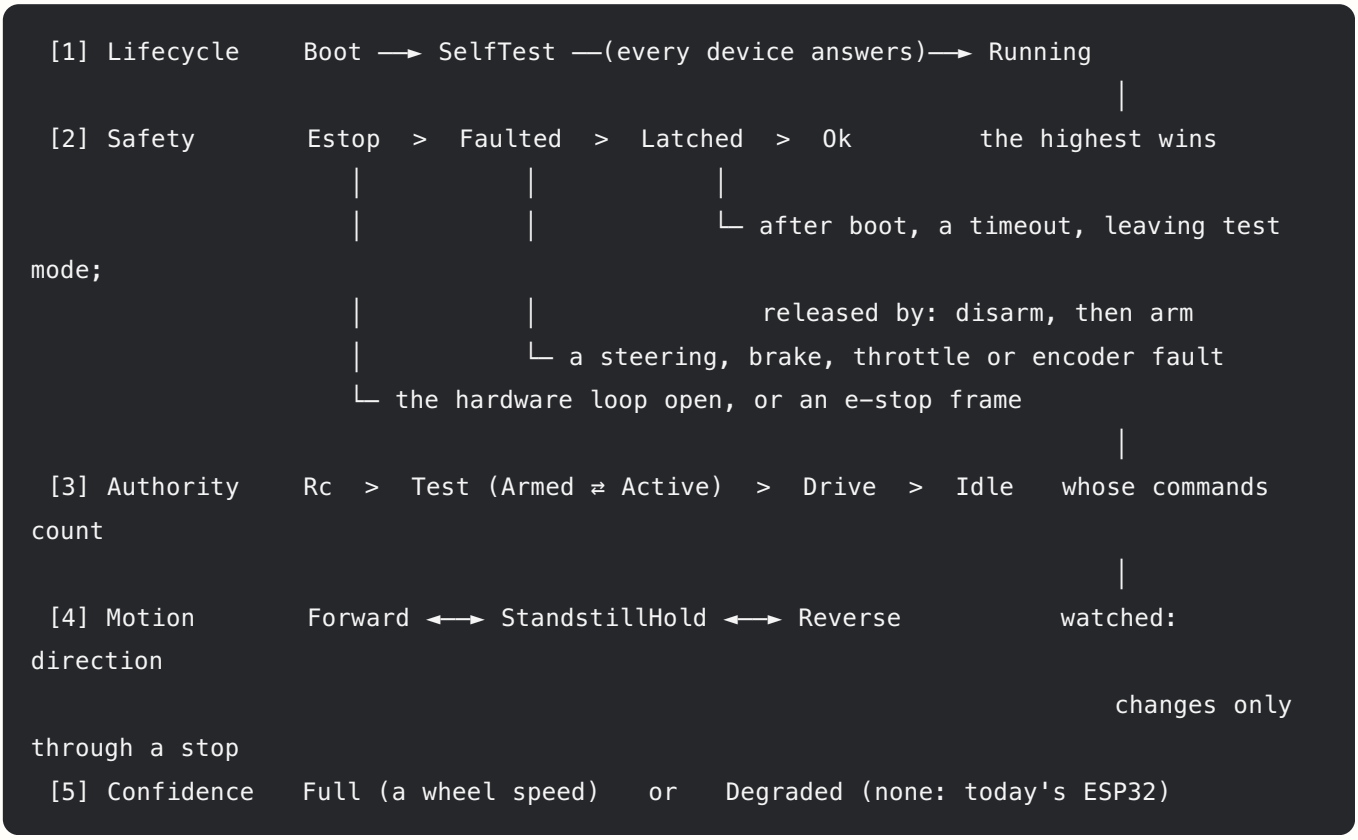
Version	Frame	Who speaks it	Use
1	13 bytes, no heartbeat	the C++ and the Rust (`--features link-v1`)	**today's car** (`frame_version: 1`)

Version	Frame	Who speaks it	Use
2	17 bytes: the Orin's counter, armed and e-stop bits, CRC-8	the Rust	the day the Rust is flashed
3	20 bytes: version 2 plus the steering calibration (limp, jog)	the Rust (reads 2 and 3 alike)	the hardware panel's STEERING CAL

Tip: the C++ can misread a version 2 stream, so `frame_version` changes only with the firmware. More: Rust firmware.

The state machine

Arsh's `fsm.rs` (`firmware/spine/crates/spine-core/src/fsm.rs`, PR #15). Five regions, each answering one question, worked out once a tick in this order.



What each state means for the actuators

State	Throttle	Brake	Steering
Safety **Estop** or **Faulted**	0	full (the stop brake)	held where it is

State	Throttle	Brake	Steering
Latched, after a timeout or fault (nobody driving)	0	full	held
Authority **Idle** (after boot)	0	0: the brake lets go (`IDLE_BRAKE`, proposed)	held
Authority **Rc** (RC direct)	from the sticks, up to the speed caps	from the speed loop; the gear switch's brake: speed 0	from the stick, up to full lock
Authority **Drive** (the computer, armed, fresh)	from the speed loop (open loop today)	from the speed loop	the command, at most `max_steer_rate`
Test, Armed	0	full	held
Test, Active	one actuator: throttle ≤ 0.15, forward only, brake off	or brake 0-1	or an angle inside the limits
Motion **StandstillHold**	0	the hold brake	as asked

Leaving test mode: above 0.5 m/s, an e-stop, a fault, RC direct, an Exit, or a request older than 100 ms. Leaving always latches: the computer must re-arm.

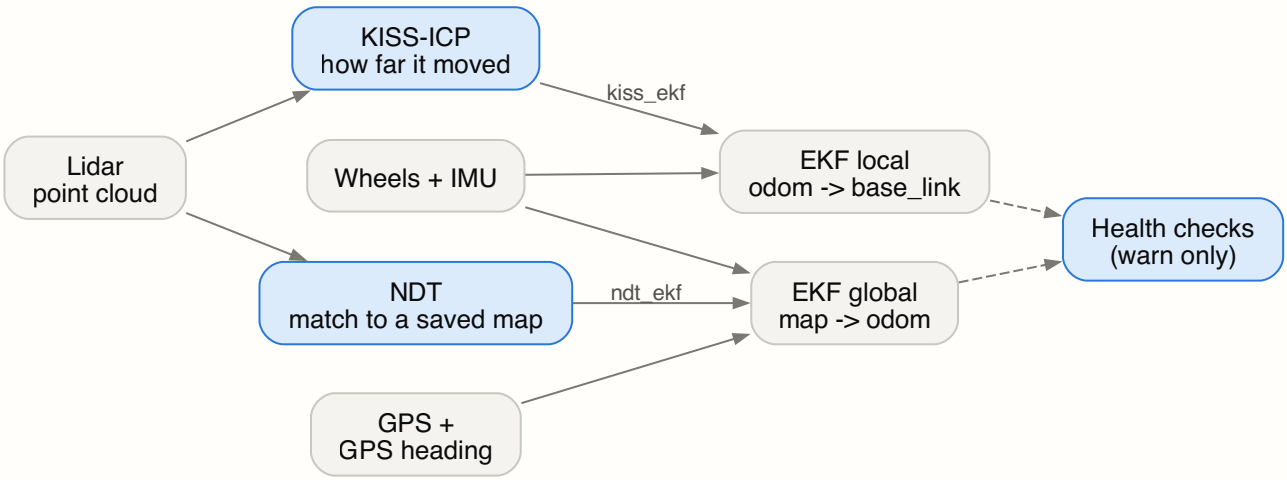
The gap

Gap	What it means	What the code does
Today's board has no wheel speed (Confidence = Degraded)	the "faster than 0.5 m/s" exit can never fire, and there's no overspeed check	the test mode **refuses a throttle test** (holds still); the brake and steering tests still work (test `without_a_wheel_speed_the_test_mode_refuses_the_throttle`)

Gap	What it means	What the code does
No way in from outside yet	no CAN frame or serial bit carries a test request	only `spine-core`'s tests reach the test mode today

Run it	`cd firmware/spine && cargo test -p spine-core` (the state machine's own tests)
How solid	built: tested on a laptop and in QEMU. Not on the car (needs A8)

Localization: where am I?



The point cloud feeds NDT and KISS-ICP (blue, trials) beside the two EKFs.

The two EKFs (the default)

What it does	`ekf_local` blends the wheels and the IMU: smooth, never jumps, drifts slowly; Nav2 steers by it. `ekf_global` adds the GPS (through `navsat_transform`) and the GPS heading: knows where on the map, may jump a little; Nav2 plans by it
Run it	in every profile: `pixi run sim`. Alone: `ros2 launch aav_localization localization.launch.py`
Tips	about 17 cm off the truth in Gazebo. Check the heading: `pixi run tools/ros.sh --local python3 tools/heading_check.py`

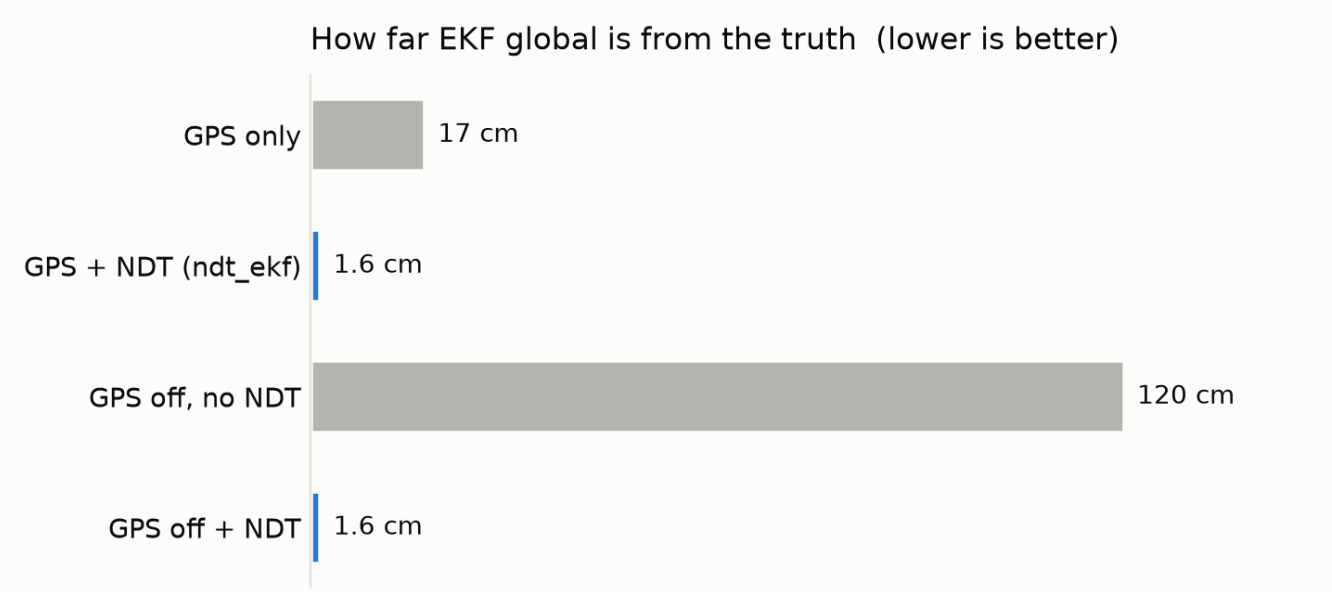
How solid	sim. Stock `robot_localization`

The sensors

Sensor	Where it comes from	Node	How solid
GPS	u-blox on the localization Pi	stock `ublox` driver, then `navsat_transform`; `gps_heading` (ours)	sim
IMU	Yahboom IMU	`aav_imu_yahboom` (Rust)	sim; the driver built
Wheels	the hardware lead's Arduino, CAN 0x200 and 0x201	`spine_bridge role:=sensors` (Rust)	built; the Arduino's frames untested

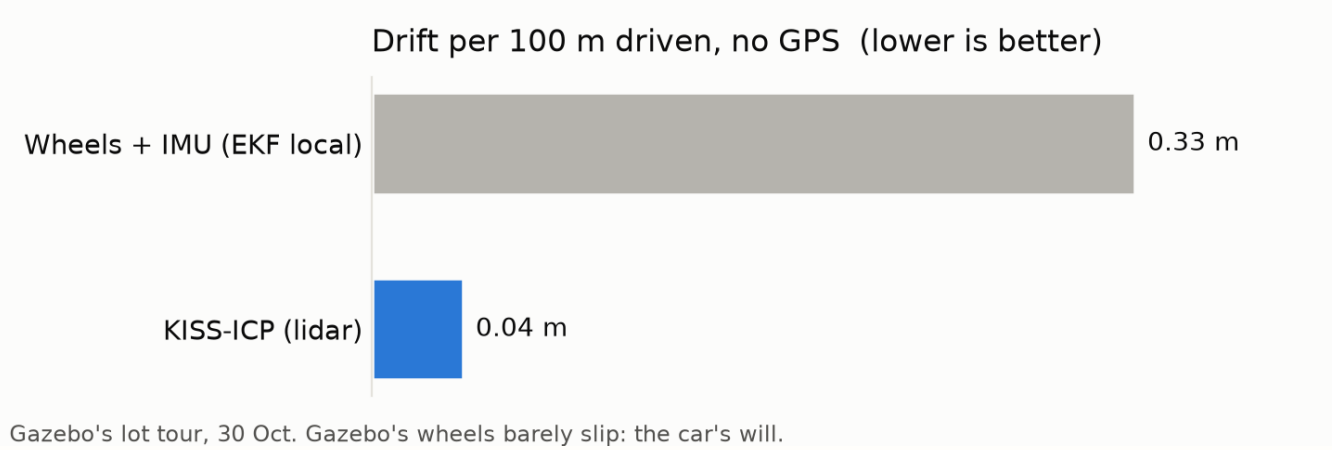
AMCL, NDT, NDT-EKF, KISS-ICP (trials)

	AMCL	NDT	NDT-EKF	KISS-ICP
What it does	many guesses; keeps those that match the walls	slides the lidar onto a saved map of the lot	NDT, and the global EKF **uses** it beside the GPS	how far the car moved, from the lidar alone (no map)
Run it	`pixi run sim world:=test_lot amcl:=true`	`pixi run sim world:=test_lot ndt:=true ndt_map:=maps/ndt/test_lot.pcd`	`... ndt_ekf:=true ndt_map:=maps/ndt/test_lot.pcd`	`pixi run sim world:=test_lot kiss_icp:=true` (`kiss_ekf:=true` : into the local EKF)
Off the truth (Gazebo)	6-11 cm	1-3 cm, 2 ms a match	the EKF 1.6 cm	0.04 m per 100 m
Tips	needs `world:=test_lot`	make the map once: `pixi run ndt-map drive` (about 6 min, not in git)	the winner, but off: the real lot needs its own map first	once: `pixi install --locked -e kiss`
How solid	trial, sim	trial, sim	trial, sim	trial, sim



Gazebo's lot tour, 30 Oct.

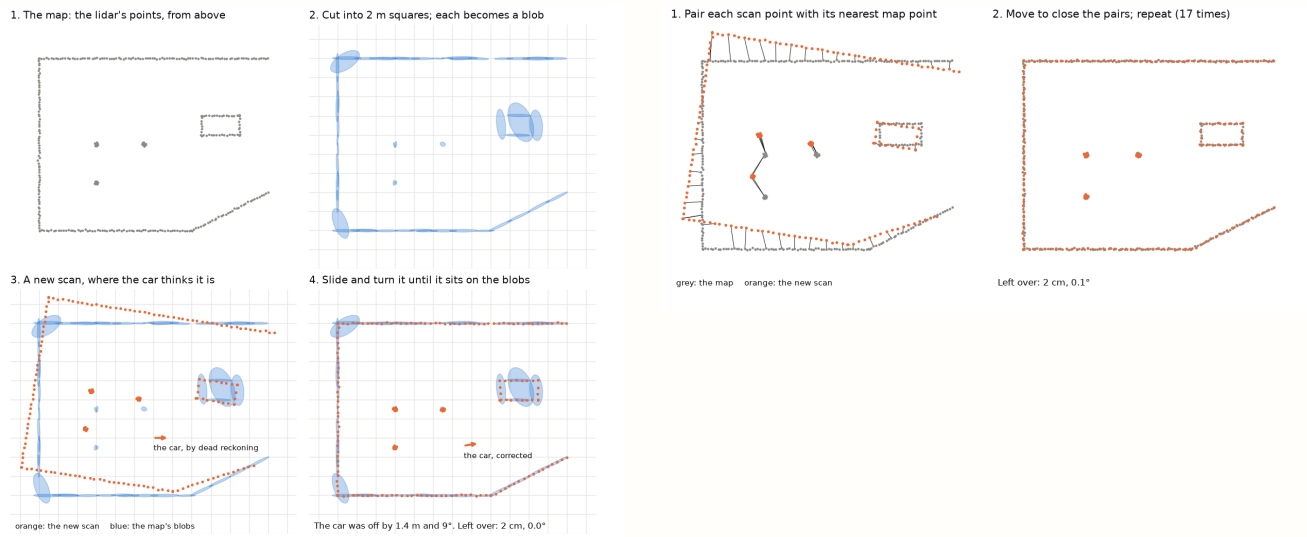
With NDT, the global EKF stays within 2 cm, with or without the GPS.



Gazebo's lot tour, 30 Oct. Gazebo's wheels barely slip: the car's will.

KISS-ICP drifts 8 times less than the wheels, with no GPS (Gazebo's wheels barely slip).

NDT	ICP (KISS-ICP's idea)



Left: NDT turns the map into blobs and slides the scan onto them. Right: ICP pairs each point with the nearest, then moves to close the pairs.

More: Trials, Learn: localization.

Why don't the two EKFs talk?

Ben, 1 Nov. **On purpose.** Checked in `ekf_local.yaml`, `ekf_global.yaml` and `localization.launch.py`: neither EKF reads the other's topic. They meet in **TF**.

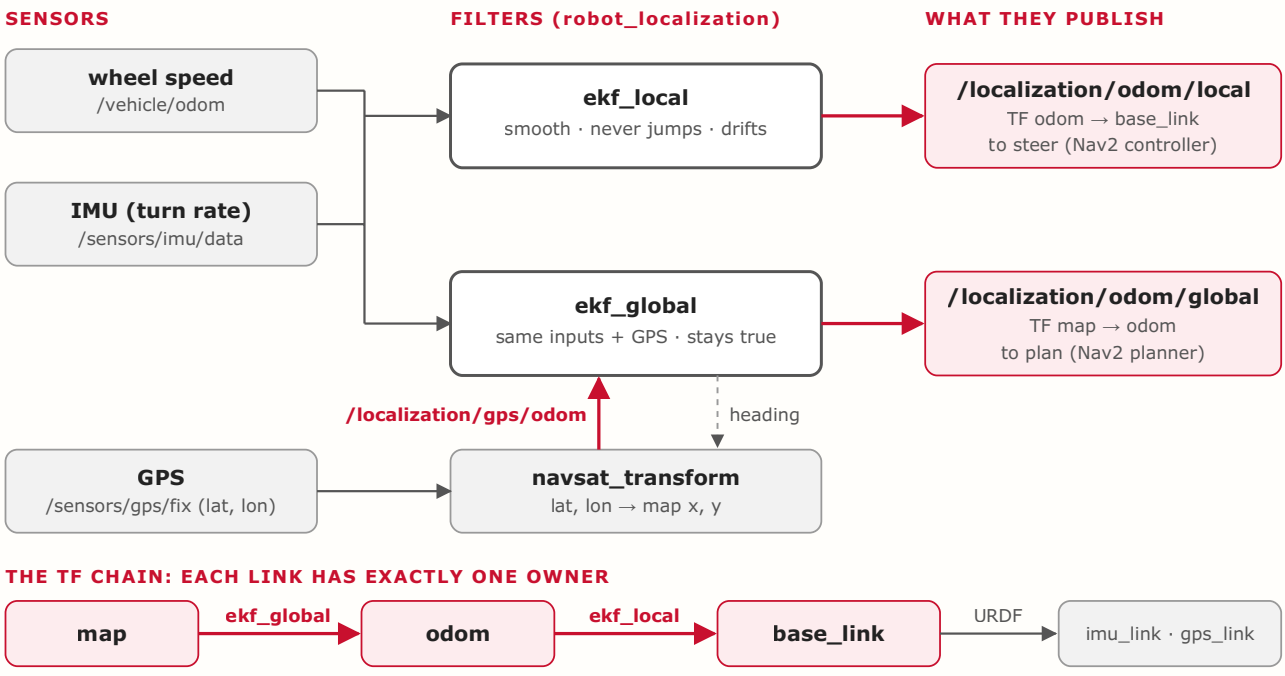
```
 wheels  /vehicle/odom ──────┬──────────────► ekf_local  ──► TF odom → base_link
(Nav2 steers by it)
 IMU     /sensors/imu/data ──┤                (world_frame: odom)
                             │                       │
                             │                       │ TF: odom → base_link
                             │                       ▼
                             └──────────────► ekf_global ──► TF map → odom
(Nav2 plans by it)
 GPS → navsat_transform → /localization/gps/odom →  (world_frame: map)
 GPS heading ──────────────────────────────────────►       │
          ▲                                               │
          └──────────── /localization/odom/global ◄───────┘   (navsat needs it to
place the GPS)
```

	`ekf_local`	`ekf_global`
Fuses	wheel speed (x, y speed, turn rate), IMU turn rate	the same, **plus** GPS x, y and the GPS heading
Its frame	`odom`	`map`
Publishes	`odom → base_link`, `/localization/odom/local`	`map → odom`, `/localization/odom/global`
Used by	Nav2's path follower	Nav2's planner, the map window

Why not feed one into the other?

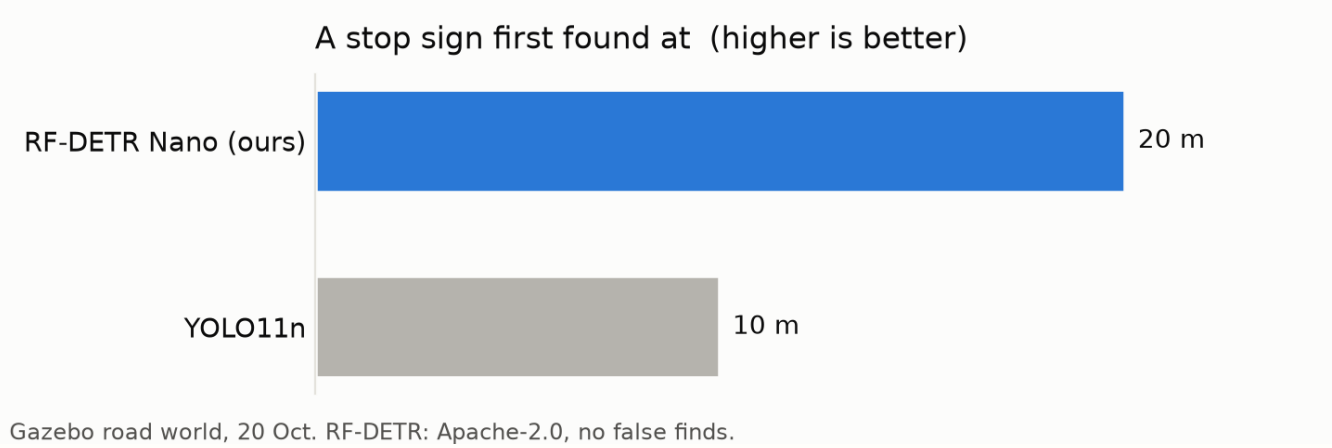
If `ekf_global` read `ekf_local`'s output	Why that's bad
the wheels and the IMU would count twice	it would be too sure of itself, and ignore the GPS

So each fuses the raw sensors itself; that's `robot_localization`'s standard setup (REP-105). `ekf_global` works out only the **correction** `map → odom`, on top of `ekf_local`'s `odom → base_link`. `rqt_graph` shows no line between them; the TF tree (`ros2 run rqt_tf_tree rqt_tf_tree`) shows the chain. More: Topic areas.



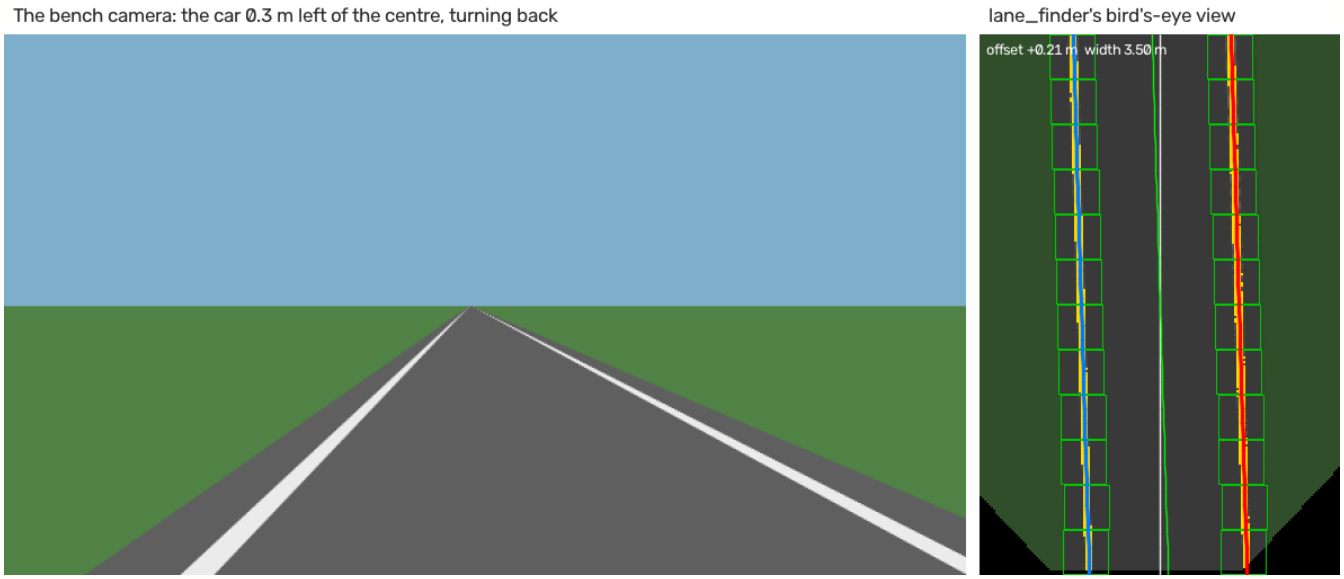
Perception: what's around?

	RF-DETR Nano and the camera stream	The cubes (voxels)
What it does	the camera Pi streams H.264 (MediaMTX); on the Orin, one `camera_stream` per camera unpacks, undistorts and runs RF-DETR Nano: objects and stop signs (Apache-2.0, not YOLO)	turns the lidar's points into 0.25 m cubes for Nav2, in place of a flat scan
Run it	`pixi run get models` once; `pixi run detect picture:=<file>`; the whole stream: `pixi run get mediamtx`, then `pixi run measure stream`	`pixi run sim world:=road voxels:=true`; see them in `demo --debug`
Tips	on the Orin it needs its TensorRT engine, built there: `python3 tools/build_engine.py`. A Mac: 27 ms from picture to detections	stopped 0.44 m before a post (the scan 0.36 m); 6 ms a cloud
How solid	sim and bench; the Orin's engine untested	trial, sim

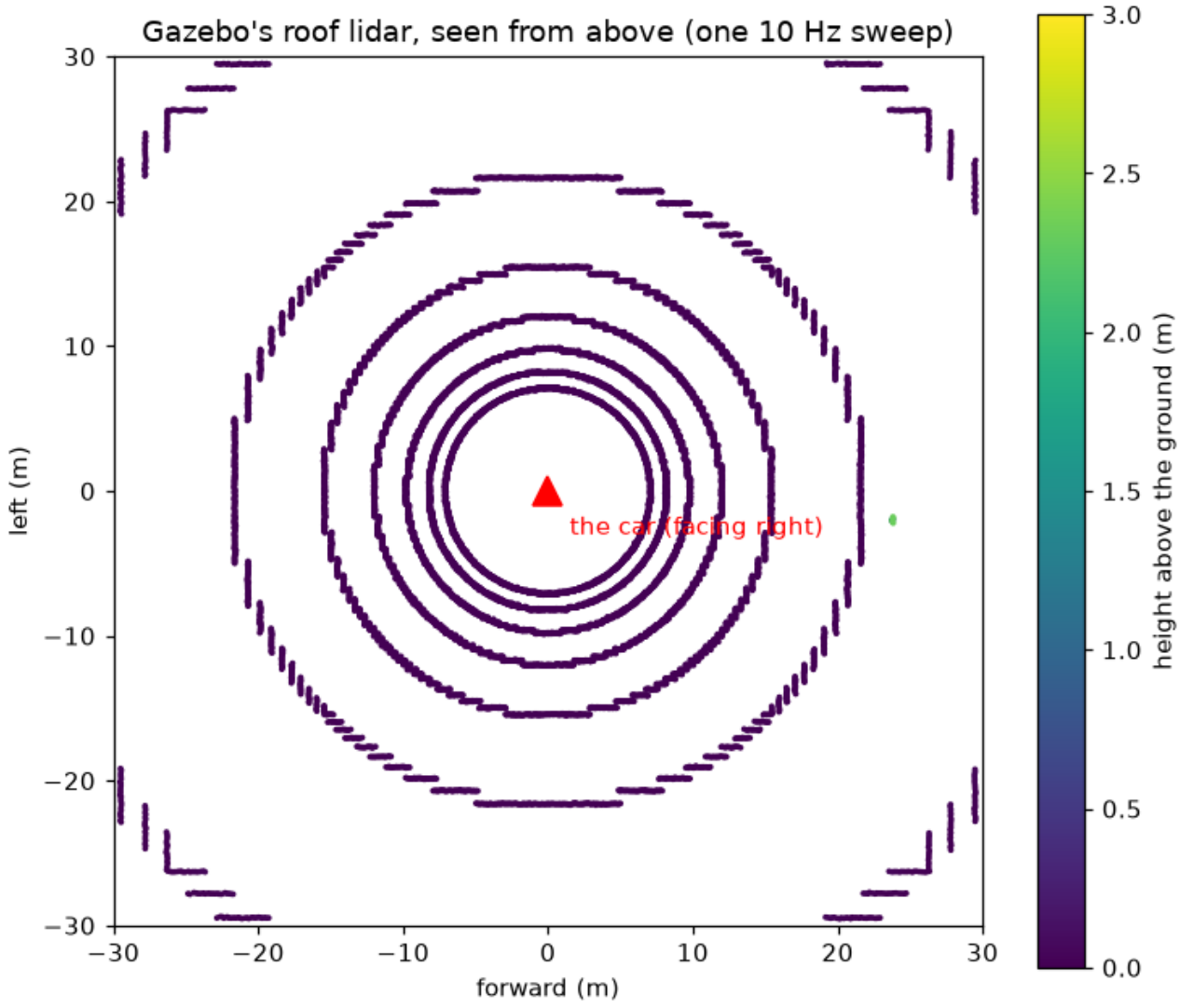


Gazebo road world, 20 Oct. RF-DETR: Apache-2.0, no false finds.

RF-DETR Nano finds a stop sign at 20 m; YOLO11n at 10 m.



The lane finder (OpenCV): the camera, and its bird's-eye view.



The roof lidar from above, one sweep: what the flat scan and the cubes are made from.

More: Perception, Camera pipeline.

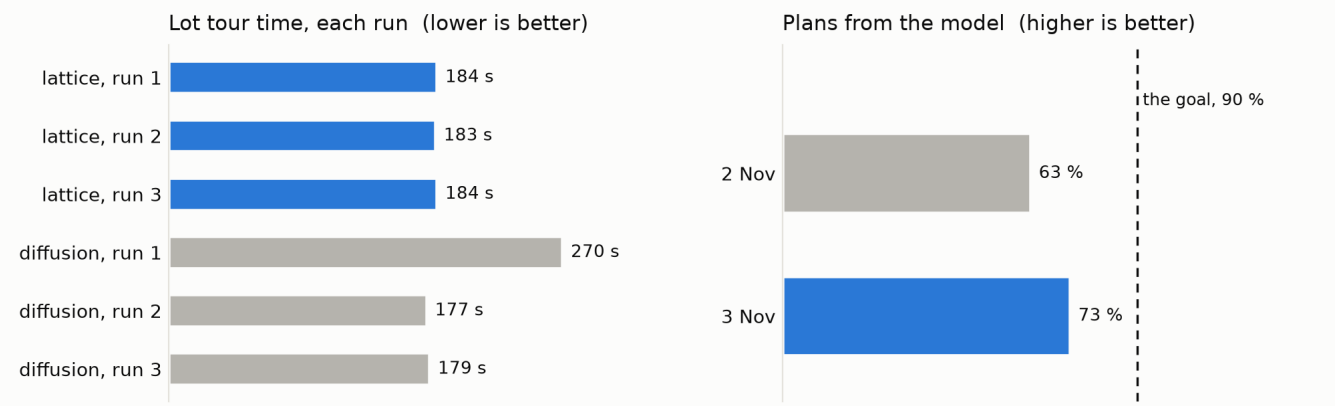
Planning: drawing the path

A **planner** draws the path; a **path follower** steers along it.

Planner	What it does	Run it	Lot tour (Gazebo)	How solid
State Lattice (default)	joins short moves the car can really make, forwards and backwards	`pixi run sim world:=test_lot`	every leg, 184 s	sim, the default
Hybrid-A* (forward only)	searches forward moves only	`... planner:=hybrid`	every leg, 213 s	sim, the default when forward only
Diffusion	a small model draws 16 paths; a guard checks each; the lattice's path if none passes or it's late	`... planner:=diffusion`	every leg; 177, 179 and 270 s; 73 % of plans from the model (3 Nov)	trial; the car refuses it
smac2d, navfn, theta_star	grid planners: ignore the turning circle	`... planner:=smac2d`	2 of 4 (smac2d)	sim only; the car refuses them

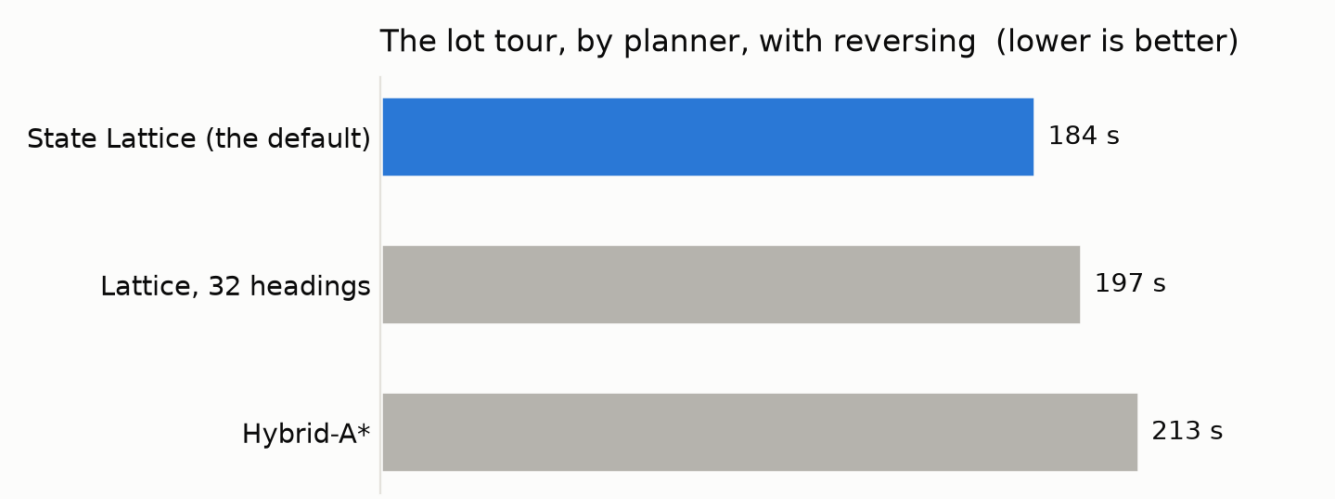
Diffusion, the tips

- Once: `pixi install --locked -e learn`, `pixi run diffusion data`, `pixi run diffusion train`.
- Not a winner yet: the design asks for 90 % from the model. The U-turn is its weakest leg (52 % from the model there), though 5 s quicker than the lattice's.
- Model and guard take 119-126 ms (mean) of a 150 ms budget beside Gazebo on a Mac: too close.



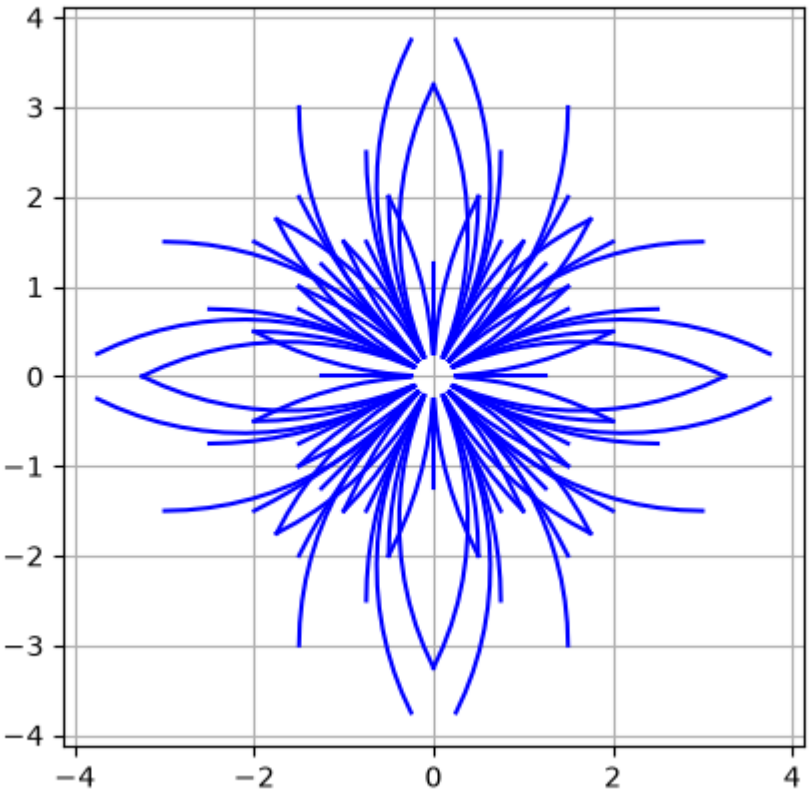
Gazebo's lot tour, reversing on, MPPI, 3 Nov; 3 runs each, 12 of 12 legs every run. Diffusion run 1: one slow home leg (149 s).

3 Nov: diffusion beat the lattice in 2 of 3 runs; one slow home leg made its first run 270 s. The model drew 73 % of the plans (2 Nov: 63 %); the goal is 90 %.



Gazebo, 30 Oct, MPPI; mean of 4 runs (lattice32: 3). Every planner finished every leg.

30 Oct: the lattice drives the lot tour 29 s quicker than Hybrid-A*.



The lattice's moves: short arcs our car can really drive, forwards and backwards.

More: Switches, Trials.

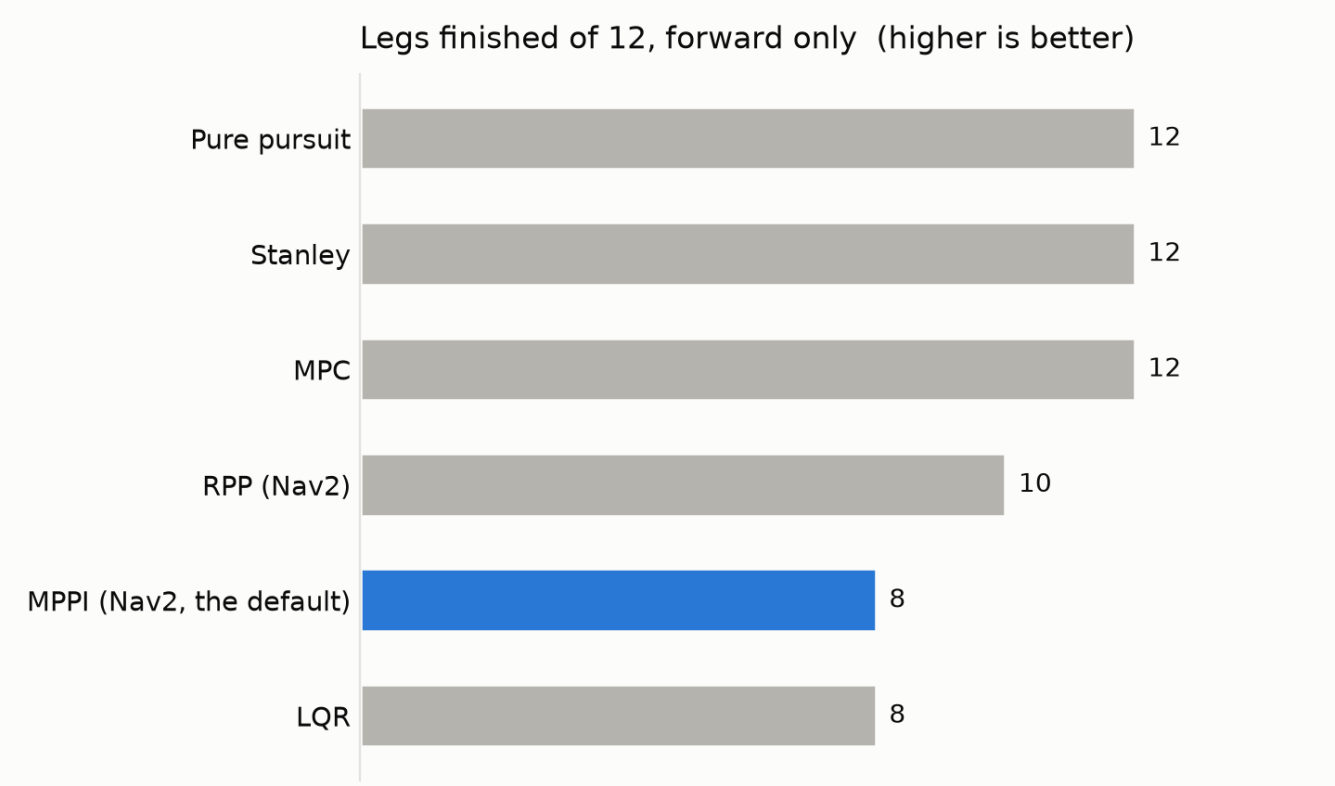
Following: steering along the path

Follower	What it does	Run it	Gazebo, 29 Oct	How solid
MPPI (default)	tries hundreds of short futures, keeps the best; reads the costmap	`pixi run sim world:=test_lot`	8-10 of 12 legs	sim, the default
RPP	aims at a point ahead, slows in bends	`... controller:=rpp`	10 of 12	trial

Follower	What it does	Run it	Gazebo, 29 Oct	How solid
pure pursuit, Stanley (ours)	aim ahead; or fix the distance and heading off the path	`... tracker:=pure_pursuit reversing:=false`	12 of 12	trial
LQR (ours)	a formula for the best steering each moment	`... tracker:=lqr reversing:=false`	10 of 12	trial
MPC (ours)	tries the next 1.5 s of steering, keeps the best	`... tracker:=mpc reversing:=false`	12 of 12	trial

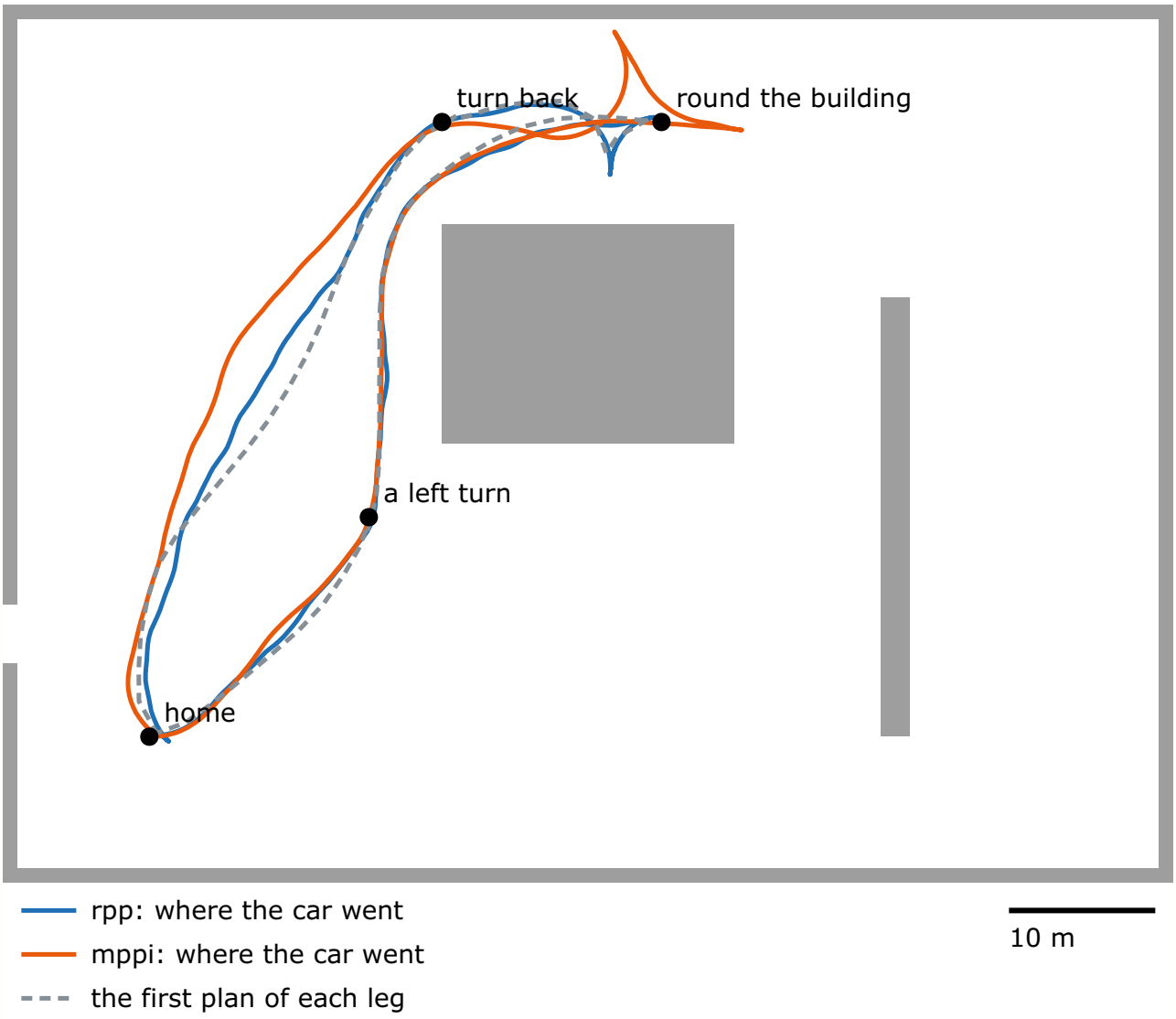
Tips

- Our trackers drive forward only: always with `reversing:=false` (`pick` does it).
- Compare them on the same routes: `pixi run measure controllers`.



Gazebo's lot, 29 Oct, 3 runs of 4 legs each. The U-turn decided it. MPPI stays the default: it reads the costmap.

29 Oct, forward only: MPPI finished the fewest, but stays the default: it reads the costmap.



Reversing allowed: both followers make the U-turn by backing up (a three-point turn); MPPI's turn is the wider.

More: Controllers, Switches.

Safety: the layers that stop the car

From the person outward. Each works even if the ones after it fail.

Layer	Stops the car when	Run it, or clear it	How solid
The Kar-Tech stop (hardware)	the e-stop person presses it: the main stop		the car's own
E-stop (`/safety/estop`)	anyone presses: the HUD, the joystick's button 2, the panel	`pixi run estop` ; `pixi run clear estop`	sim
Hold-to-drive	the joystick's trigger or the panel's Space is let go		sim
ego's timeout	no command for 0.3 s		sim
The output mux	any data isn't the real car's (sim, bench, replay on the network): sends nothing, latched (`STOP`)	`/safety/output_mux/state` ; `pixi run clear gate`	sim
The sensor watchdog (Nav2 and the AI only)	a sensor or an EKF goes quiet, sends NaN, freezes or jumps; the HUD dies	try: `pixi run sim-fault gps stop` ; `pixi run clear watchdog`	sim
collision_monitor (Nav2 only)	lidar or radar points in the stop zone ahead	try: `pixi run ghost ahead 10`	sim
The geofence (programs only, never people)	the stopping path would leave the fence (30 m round the start)	`pixi run bench-check --geofence` ; `pixi run clear geofence`	bench
The firmware's watchdog	100 ms without a new heartbeat	`pixi run esp qemu-check`	QEMU only

Tips

- `clear` is always a person's job: nothing clears itself after a STOP.
- With `frame_version: 1` (today's C++ ESP32), the bridge locks every program out: only people and the actuator tests drive.

More: How it fits, safety layers, Sources and targets, The geofence.

At the car

car_setup.sh: one computer at a time

What it does	on one car computer: back up, upgrade, install our code, start it at boot, check. It only prints the steps unless given `--apply` and the role typed in
Run it	copy it over (`scp tools/car_setup.sh <user>@<address>:`), log in, `tmux new -s setup` , then `bash car_setup.sh motor_pi` (a dry run) and `bash car_setup.sh motor_pi backup --apply`
Tips	a STOP line ends the run before anything changes: a pending reboot, the wrong hostname, an upgrade too early. `motor_pi` and `localization` refuse to upgrade on purpose until their way back is proven
How solid	dry run passes for all six roles. `--apply` untested on the car

More: Car setup.

Car day, in two days

Day	What	Can you undo it?
Day A	back up every machine (checked); the steering's ends (Way A); the brake on stands; the Rust ESP32 and the joystick on stands; walking pace once a named person signs; the car tests and the map	yes: nothing destroyed
Day B (later)	the Pis to Ubuntu 24.04; the Orin reflashed to JetPack 7.2; the lab desktop	only from the backups, so a restore is proven first

Role	Does
Hardware lead	decides on the hardware; OKs the ESP32 flash
E-stop person	holds the Kar-Tech stop, always
Safety driver	in the seat
Laptop person	runs the commands

Role	Does
Approver	signs the steps you can't undo

Run it	the night before: `pixi run car-day-check` until it says 0 missing; `pixi run car-day-build`
Tips	every step you can't undo has a sign-off line (who, time). Record all day
How solid	written. The runbook: Car day

The measurements

What it does	every car number lives in one file, `ros/bringup/aav_description/config/vehicle.yaml` (many marked GUESS). Each car test fixes one: lock to lock, a fixed-steering circle, 10 m by tape (the wheel size), a speed step, a brake, a coast-down
Run it	the sheet: Measure the car (`pixi run python3 tools/measure_map.py` for the tape route); the tests: `pixi run actuator-test --list`; the order: Know the car
Tips	steering first, then the wheels, then speed and brake. The wheel size is 0.36 m or 0.42 m today: a 17 % speed error until it's taped. See a change's effect: `pixi run python3 tools/vehicle_model.py --set <key>=<value>`
How solid	written; the tests run in Gazebo (`actuator-test <test> --sim`)

Recording

What it does	records a drive to `bags/<date>_<name>/` (MCAP): raw inputs, commands, what the stack decided. No camera pictures
Run it	`tools/record.sh --lidar <name>` (on the car: `~/aav-jazzy/aav.sh tools/record.sh --lidar <name>`)
Tips	`--lidar` is about 9 MB/s (10 min is about 5 GB), but the same drives build the lot's NDT map: `pixi run ndt-map build`. Bags stay out of git
How solid	sim

Tools

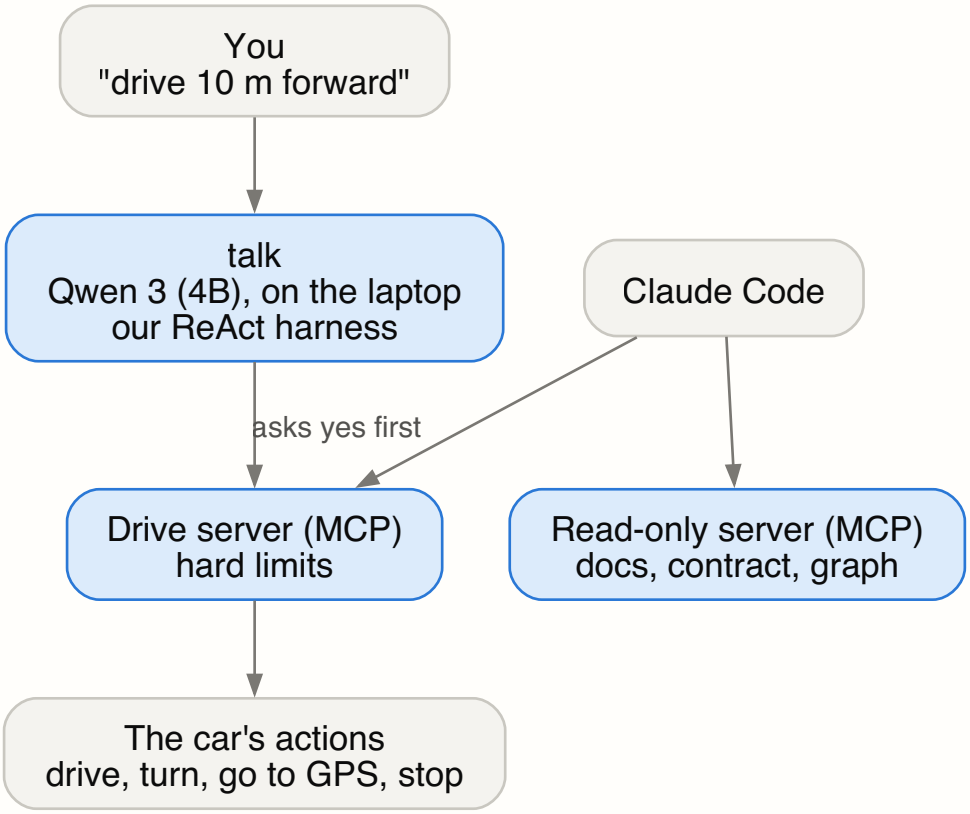
Foxglove and Lichtblick

What it does	watch the car or a simulator from any laptop, with no ROS on it: one app, one WebSocket (`foxglove_bridge`, port 8765)
Run it	`pixi run sim foxglove:=true` (on the car it runs already); in the app: Foxglove WebSocket `ws://localhost:8765` (or the Orin's address); import a layout from `tools/foxglove/`
Tips	Lichtblick is our pick (open source, no account). Its Mac app isn't notarized: right-click → Open the first time. Layouts: driving, sensors health, map, cockpit
How solid	sim. On the car: watch only

More: Watch from a laptop.

The MCP server

What it does	lets an AI app such as Claude Code read the docs, the topic contract and the running ROS graph. **It only reads**: it can't move the car
Run it	start Claude Code in the repo folder: it loads the server (`.mcp.json`). By hand: `pixi run -e mcp mcp`
Tips	it sees the ROS network it was started on: the car's, if your laptop is on it
How solid	works; `tools/tests/test_mcp_parts.py` checks it never writes



Two ways to talk to the car: `pixi run talk` (Qwen, on the laptop, asks yes first) and Claude Code through the drive server. Both go through the drive server's hard limits.

More: AI apps (MCP).

The commands

89 commands became 45. Each is `pixi run <command>`; a family is one command and a word after it (`pixi run get models`). The source, with more words: README's "Every command".

Group	Command	One line
Every day	`build`	compiles our ROS packages
	`test`	every test and the style check, this computer only
	`lint`	checks our Python against ROS's style rules
	`docs`	rebuilds the web pages and the PDFs

Group	Command	One line
Try it, no car	`turtle`	turtlesim plays the car
	`keyboard`	drive from the keyboard: `--local` with `turtle` or `bench`, `--sim` with `sim`
	`sim`	Gazebo plays the car (`world:=road`, `hud:=true`...)
	`demo`	Gazebo, RViz, the HUD and the map at once; `--debug` adds the debug screen
	`pick`	a menu of the switches: prints the command and runs it; `pick last` again
	`bench`	the real car's software, stand-ins for the hardware
	`view`	the car's model in RViz
	`carla-car`	CARLA plays the car: `host:=<the CARLA computer>`
	`drive-to-point`	drive to one point and stop smoothly: `--sim --ahead 20`
	`ghost`	a pretend person in Nav2's view: `ghost ahead 10`
	`sim-fault`	make a Gazebo sensor fail: `sim-fault gps stop`, then `gps ok`
Screens (`--demo`: no ROS)	`hud`	the in-car screen
	`map`	the big screen's map window
	`debug-hud`	the debug screen: cubes, localizers, every sensor
	`hw-panel`	the hardware panel: every sensor, simple buttons, hold Space to drive
	`rviz`	RViz with the driving layout
	`graph`	a live picture of the nodes and topics
Check it works	`check`	the running graph against the contract: `check sim` (or `turtle`, `road`, `spine`, `car`, `replay`, `bench`)
	`bench-check`	bench mode, the whole graph tested (2 min)

Group	Command	One line
	`sim-check`	Gazebo's road world, headless, the whole graph tested (2 min)
	`measure`	`controllers`, `timing`, `stream`, `atob <bag>`, `bandwidth`
	`diffusion`	the diffusion trial: `data`, `train`, `score`
	`ndt-map`	NDT's map of the test lot: `drive` or `build`
At the car	`car`	the real car: `car orin` (or `motor_pi`, `laptop`, `all`)
	`estop`	block every driver: the car brakes
	`clear`	a person clears a latch: `estop`, `gate`, `geofence`, `watchdog`
	`actuator-test`	the actuator tests, on stands and in a lot: `--list`
	`calibrate-camera`	calibrate a streamed camera
	`doctor`	can this computer join the car's ROS network?
Before car day	`car-day-check`	is this laptop ready? Until 0 missing
	`car-day-build`	builds what the laptop takes to the car
	`car-kit`	our Jazzy for the car's computers, on a USB stick
	`get`	downloads what git doesn't keep: `vendor`, `models`, `mediamtx`, `places`, `street-map`, `carla-osm`, `routes`, `all`
Firmware (bench only)	`pico`	our Rust Pico 2 boards: `pico <board> <tools, build or flash>`
	`esp`	the Rust ESP32: `tools`, `build`, `flash`, `qemu-tools`, `qemu-check`
	`-e firmware legacy-esp32`	today's C++ ESP32 firmware (PlatformIO)
Perception, AI, maps	`detect`	the object detector on a picture or the cameras
	`laptop-bench`	your laptop's detector against bench mode

Group	Command	One line
	`talk`	tell the car what to do in plain English (turtle, sim or bench)
	`-e mcp mcp`, `-e mcp drive-server`	the MCP servers: read-only, and the drive tools
	`-e routes router`	the street router (needs `get routes`)

Old	New
`hud-demo`, `debugdemo`	`hud --demo`, `demo --debug`
`estop-clear`, `gate-clear`	`clear estop`, `clear gate`
`esp-build`, `spine-flash`	`esp build`, `pico spine flash`
`get-models`, `vendor`	`get models`, `get vendor`
`controller-compare`, `timing-test`	`measure controllers`, `measure timing`

Tip: a family run with no word lists its choices (`pixi run get`).

Which parts are Python, and is that OK?

Ben, 1 Nov: "I don't want performance critical code running in python".

The short answer: the hard timing isn't in Python. The 100 ms brake lives in the Rust firmware. In the Python nodes, the heavy maths runs in C, C++ or Rust libraries (NumPy, ONNX Runtime, TensorRT, GStreamer, KISS-ICP's core). Every number below fits its budget on a Mac. **Nothing is measured on the Orin yet** (TODO 76: `pixi run measure timing` on car day).

Node (on the car)	Language	Must keep up with	Measured (where)	0.3
The ESP32 firmware (`spine-core`)	**Rust**	100 ms watchdog	brakes 111-120 ms after the heartbeat stops (QEMU)	stays
`spine_bridge` (CAN)	**Rust**	50 Hz		stays

Node (on the car)	Language	Must keep up with	Measured (where)	0.3
IMU driver, radar clusters, NDT, the screens	**Rust**	50 Hz; 10 Hz clouds	NDT 2 ms a match, worst 6.3 (Gazebo)	stay
twist_mux, the EKFs, `navsat_transform`, Nav2, the u-blox and lidar drivers	**C++** (stock)	30-50 Hz	the lattice 4-11 ms a plan, mean (Gazebo)	stay
ego	Python	50 Hz (20 ms)	not measured: a few sums a tick	a candidate, once the car drives
output mux, the heartbeat	Python	every command, 50 Hz	not measured; the ESP32 brakes if it stalls	stays: a stall is caught downstream
`legacy_uart_bridge` (Motor Pi)	Python	50 Hz frames	not measured; the same catch	stays
`joy_alive`	Python	20 Hz; decides over 0.5-2 s		stays
sensor watchdog, geofence, stop hold	Python	10 Hz		stay
`camera_stream` + RF-DETR	Python glue; GStreamer and TensorRT do the work	15 fps (67 ms)	27 ms picture to detections (Mac, Core ML: 10 ms a picture)	stays; the Orin to measure
flat scan (`aav_lidar`)	Python, NumPy	10 Hz clouds (100 ms)	0.18 ms a cloud	stays
the cubes (`aav_voxels`, trial)	Python, NumPy	10 Hz clouds (100 ms)	6.1 ms mean, 12.7 worst	**Rust first, if the Orin takes over 20 ms** (TODO 76)

Node (on the car)	Language	Must keep up with	Measured (where)	0.3
KISS-ICP (trial)	Python node; C++ core	10 Hz	about 1 ms a scan (Mac, 4 threads)	stays
our trackers (`aav_track`, trial)	Python, SciPy	20 Hz (50 ms)	MPC 2-3.5 ms mean, 16 ms worst; 21-32 % CPU, mostly rclpy	Rust only if one wins
the diffusion planner (trial)	Python node; the model in ONNX Runtime (C++)	a plan in 150 ms	the model 75 ms (Mac CPU); the guard 19 ms in NumPy; both 126-145 ms beside Gazebo	**the guard to Rust** (`diffusion-guard`, in its design)

Is it OK? For 0.2, yes: nothing that must react in under 100 ms depends on Python alone, and the Python trials are off on the car. The two to watch are the diffusion planner (too close to its 150 ms) and the cubes (if the Orin is slow).

More: Languages.

Kept for 0.3 and later

Every idea is kept (Ben, 1 Nov: "I do like the 23 ideas? lets keep them around").

Idea	State
Car day itself (0.3 = it drove)	runbook ready
One screen per concept: **Drive** (Gazebo, Nav2, the cubes, the algorithms) and **Hardware**	Ben, 3 Nov: TODO 90
The diffusion planner: 90 % of plans from the model, the U-turn, the guard in Rust, the Orin	trial in 0.2
The talking model: explains, never drives	idea
The little robot on the in-car HUD	idea

Idea	State
Calm colours	waits on Ben
The cockpit's algorithm switch; one screen from the cockpit and the debug screen	in 0.2: the cockpit shows which algorithms run (3 Nov); the switch and the debug screen's views for 0.3
More hardware panel pages: measurements, joystick, the car tests	idea
Python speed on the Orin (the cubes to Rust if too slow)	measure on car day
Brake and throttle: measure, then set	after the steering
Day B: Ubuntu 24.04 on the Pis, JetPack 7.2 on the Orin	after a restore is proven

The list: TODO, "0.3 and later".

How solid is 0.2?

Solid	every test green; every profile runs in Gazebo; the screens seen live against it
Untested	anything on the real car: the steering page, `car_setup.sh --apply`, the boot service, the joystick path, the Rust ESP32
Open	the joystick-alive check on the real pad (A9b), the throttle cap's number, the people, the hardware lead's answers

The checks run for this page

Run on Ben's Mac, 2 November:

Check	Result	What's left
`pixi run test`	✅ 1516 tests, 0 failures	nothing

Check	Result	What's left
`bash tools/car_setup.sh <role>`, all six roles (dry run)	✅ no STOP except the two planned refusals (`motor_pi` and `localization` won't upgrade yet)	nothing
`pixi run car-day-check`	⬜ 7 OK, **11 missing**, 7 to tick by hand	mostly downloads not yet on this Mac: `get models`, `get mediamtx`, `esp tools`, `esp build`, `esp qemu-tools`, `esp qemu-check`, `car-day-build` (the C++ way back), `docs` (the PDF)
`pixi run esp qemu-check`	⬜ skipped: the ESP32's Rust and QEMU aren't installed here	`pixi run esp tools`, then `esp qemu-tools`, then run it again

The ESP32 tools install Espressif's Rust on the laptop, so they wait for Ben's OK.